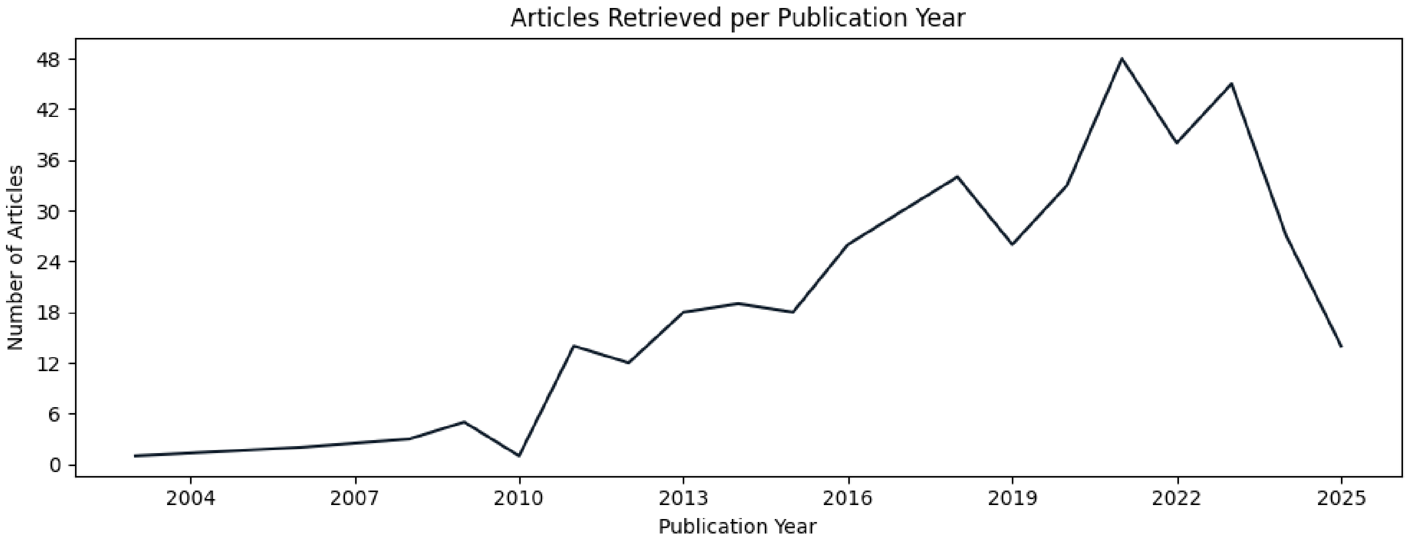
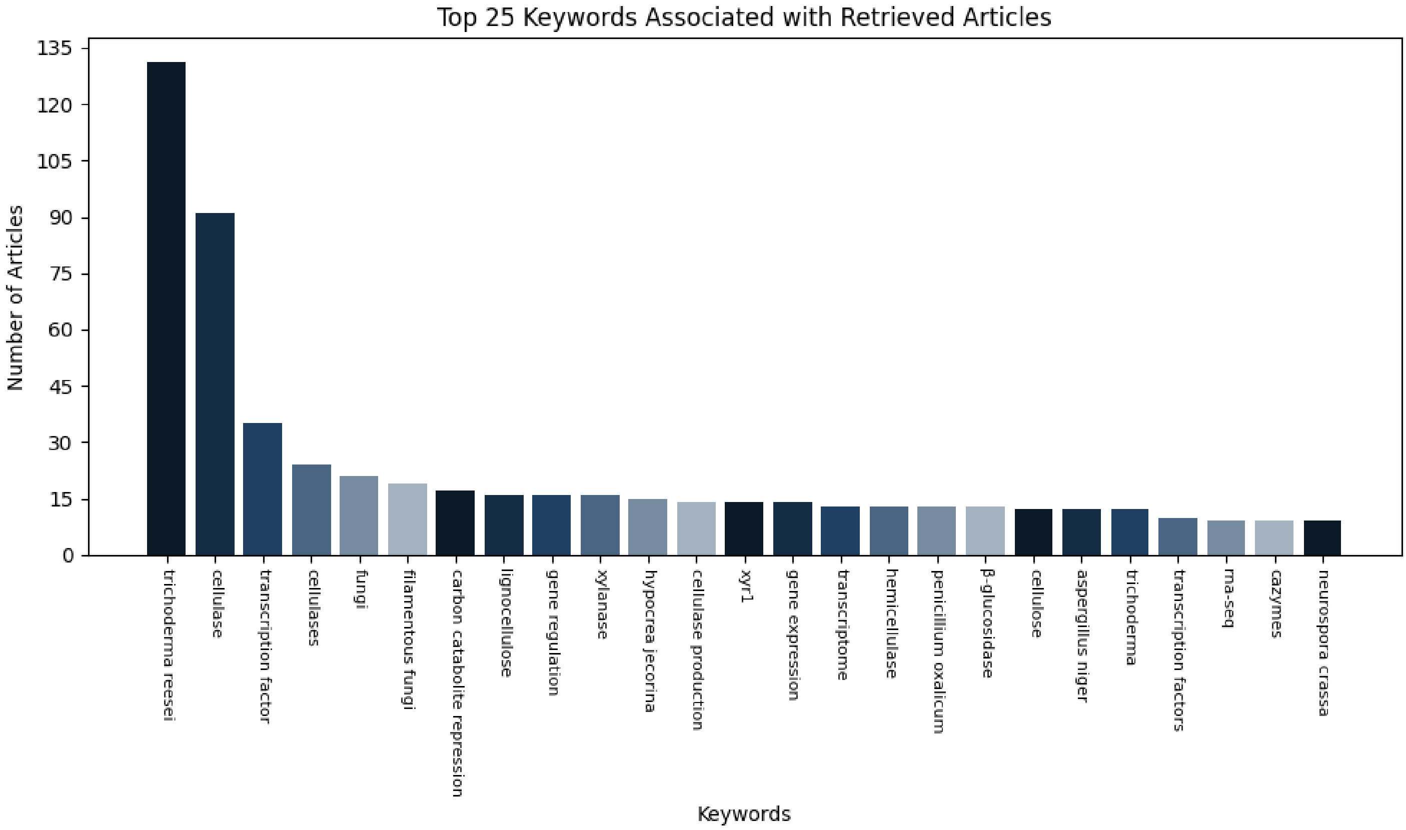
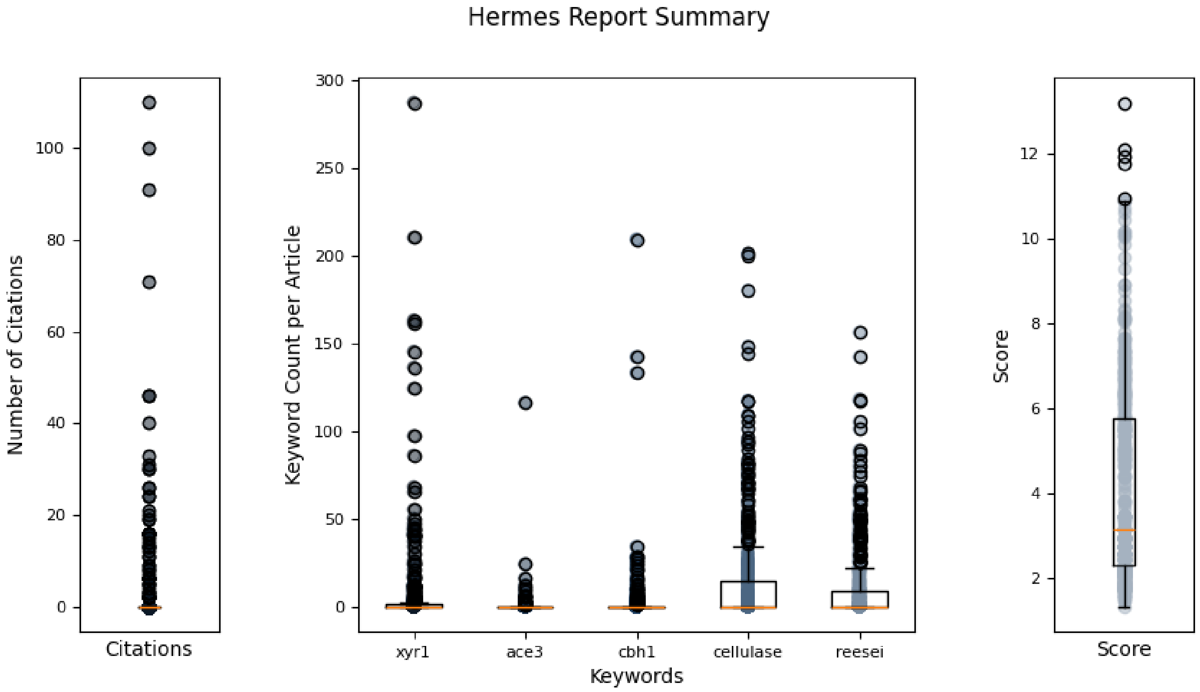
Literature Mining Report

Query Statistics

Query date: 2025-08-14
Query terms: "xyr1 cellulase production trichoderma reesei"
Articles parsed/hits: 416/416 (100.00%)
Scoring keywords: xyr1, ace3, cbh1, cellulase, reesei







Query Results

PMID	Year	Journal	Authors	Title	Citations	Score
22554051	2012	Molecular Micro	Seiboth, Bernha et al.	the putative protein methyltransferase lae1 c...	100	13.20
25909478	2015	BMC Genomics	Lichius, Alexan et al.	genome sequencing of the trichoderma reesei q...	24	12.11
32877410	2020	PLoS Genetics	Zheng, Fanglin et al.	trichoderma reesei xyr1 activates cellulase g...	19	11.93
36376415	2022	Scientific Repo	Arai, Toshiharu et al.	inducer-free cellulase production system base...	6	11.77
31077201	2019	Microbial Cell	Liu, Pei et al.	enhancement of cellulase production in tricho...	16	10.96
21980519	2011	PLoS ONE	Sun, Jianping et al.	identification of the cre-1 cellulolytic regu...	91	10.88
36732590	2023	Scientific Repo	Schalamun, Miri et al.	mapkinases regulate secondary metabolism, sex...	11	10.87
29619080	2018	Biotechnology f	Nogueira, Karol et al.	characterization of a novel sugar transporter...	30	10.72
23152805	2012	PLoS ONE	Zhang, Jiwei et al.	ras gtpases modulate morphogenesis, sporulati...	16	10.62
36528622	2022	Biotechnology f	Shen, Linjing et al.	tailoring the expression of xyr1 leads to eff...	3	10.45
32292397	2020	Frontiers in Mi	Li, Cheng-Xi et al.	weighted gene co-expression network analysis ...	13	10.18
28066400	2016	Frontiers in Mi	Li, Yanan et al.	the different roles of penicillium oxalicum l...	21	10.15
29860590	2018	Applied Microbi	Kluge, Janina et al.	inducible promoters and functional genomic ap...	40	10.11
32760377	2020	Frontiers in Mi	Li, Jia-Xiang et al.	diversity of cellulase-producing filamentous ...	9	10.06
27199949	2016	Frontiers in Mi	Wang, Min et al.	exploring the synergy between cellobiose dehy...	12	10.03
27621727	2016	Frontiers in Mi	Qian, Yuanchao et al.	characterization and strain improvement of a ...	11	9.87
24360128	2013	AMB Express	Fujii, Tatsuya et al.	enhancing cellulase and hemicellulase product...	14	9.57
28330286	2016	3 Biotech	He, Ronglin et al.	effect of highly branched hyphal morphology o...	13	9.28
39628517	2023	Engineering Mic	Wang, Yubo et al.	constitutive overexpression of cellobiohydrol...	2	8.91
35317826	2022	Microbial Cell	Mattam, Anu Jos et al.	factors regulating cellulolytic gene expressi...	0	8.91
21609467	2011	Microbial Cell	Limn, M Carmen et al.	the effects of disruption of phosphoglucose i...	6	8.78
32765463	2020	Frontiers in Mi	Jiang, Xianzhan et al.	improved production of majority cellulases in...	0	8.54
39109875	2024	Applied and Env	Chen, Siqiao et al.	genus-wide analysis of trichoderma antagonism...	0	8.33
38650205	2021	Bioresources an	Yan, Su et al.	from induction to secretion: a complicated ro...	0	8.32
38548869	2024	Communications	Zhu, Zhihua et al.	the protein methyltransferase trsam inhibits ...	0	8.18

Query Result 1/25

The putative protein methyltransferase LAE1 controls cellulase gene expression in Trichoderma reesei

Seiboth, Bernhard; Karimi, Razieh Aghcheh; Phatale, Pallavi A; Linke, Rita; Hartl, Lukas; Sauer, Dominik G; Smith, Kristina M; Baker, Scott E; Freitag, Michael; Kubicek, Christian P

Molecular Microbiology

Year: 2012 PMID: 22554051 doi: 10.1111/j.1365-2958.2012.08083.x Citations: 100 Score: 13.20

Keywords hits: xyr1: 24, ace3: , cbh1: , cellulase: 89, reesei: 8

Abstract:
Trichoderma reesei is an industrial producer of enzymes that degrade lignocellulosic polysaccharides to soluble monomers, which can be fermented to biofuels. Here we show that the expression of genes for lignocellulose degradation are controlled by the orthologous T. reesei protein methyltransferase LAE1. In a lae1 deletion mutant we observed a complete loss of expression of all seven cellulases, auxiliary factors for cellulose degradation, -glucosidases and xylanases were no longer expressed. Conversely, enhanced expression of lae1 resulted in significantly increased cellulase gene transcription. Lae1-modulated cellulase gene expression was dependent on the function of the general cellulase regulator XYR1, but also xyr1 expression was LAE1-dependent. LAE1 was also essential for conidiation of T. reesei. Chromatin immunoprecipitation followed by high-throughput sequencing (ChIP-seq) showed that lae1 expression was not obviously correlated with H3K4 di- or trimethylation (indicative of active transcription) or H3K9 trimethylation (typical for heterochromatin regions) in CAZyme coding regions, suggesting that LAE1 does not affect CAZyme gene expression by directly modulating H3K4 or H3K9 methylation. Our data demonstrate that the putative protein methyltransferase LAE1 is essential for cellulase gene expression in T. reesei through mechanisms that remain to be identified.

Summary:
Interestingly, 13 of these 28 clusters were found in areas not previously predicted as CAZyme gene clusters. Table 2 Clusters of expressed genes affected in T. reeseilae1. LAE1 dependence of cellulase gene expression is inducer independent In order to confirm the results from microarray analysis, we used cel7 A and cel6 A as cellulase model genes and tested their expression in the parent strain and in the lae1 strain by q RT-PCR. The comparatively smaller effect of lae1 overexpression on cellulose may be due to the fact that cellulases are required for growth, and their overexpression may lead to an excess of cellobiose and glucose formed that in turn induces carbon catabolite repression of cellulase expression. reesei QM 9414 and several mutant strains bearing additional copies of the tef1: lae1 gene construct on lactose. A comparison with the CAZyme genes that were downregulated in the lae1 strain showed that 27 genes were consistently affected in both strains, i. e. exhibited strongly reduced expression in the lae1 strain but enhanced expression in the tef1: lae1 strain (Fig. We must also note that although Ch IP analysis of heterochromatic marks in A. nidulans lae A strains had revealed a dramatic increase in H3 K9me3 and Hep A binding at secondary metabolism clusters a direct effect of Lae A on histone modification still awaits biochemical evidence. While the precise mechanism of action of LAE1 remains to be identified, we speculate that it could be related to the linkage between asexual sporulation and CAZyme gene transcription in T. reesei: as we have shown in this article, one of the most striking phenotypes of lae1-mutants in T. reesei is the almost complete absence of sporulation in the lae1 and hypersporulation in tef1: lae1 strains, a phenomenon already observed in some other fungi. Asexual sporulation triggers massive CAZyme gene expression in T. reesei in an inducer-independent but XYR1-dependent way, and sporulation is commonly observed at later stages of cellulase formation on lactose which was absent from the lae1 strain and increased in the tef1: lae1 mutant in this study.

Mentioned biomedical entities:
Genes: Bok, CIP1, Cellulase, GH, HP1, HepA, LAE1, LaeA, XYN1, XYN2, XYN3, XYR1, cel12A, cel1A, cel5A, cel6A, cel6a, cel7B, cellulase, cip1, hH4-1, histone, lae1, pki1, tef1, xyn1, xyn3, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Chaetomium, E., F) lactulose, NIH, Protoplast, Pyrenophora tritici repentis, TU-6, Trichoderma, cells, cellulose, ku7, plasmid, see, tef1:lae1 strains, wild-type

Organisms: A. fumigatus, E. coli hygromycin B phosphotransferase, Escherichia coli JM19, H Minus, SV Total RNA, wild-type QM 9414

Tissues: N/A

Pathways: N/A

Query Result 2/25

Genome sequencing of the Trichoderma reesei QM9136 mutant identifies a truncation of the transcriptional regulator XYR1 as the cause for its cellulase-negative phenotype

Lichius, Alexander; Bidard, Frdrique; Buchholz, Franziska; Le Crom, Stphane; Martin, Joel; Schackwitz, Wendy; Austerlitz, Tina; Grigoriev, Igor V; Baker, Scott E; Margeot, Antoine; Seiboth, Bernhard; Kubicek, Christian P

BMC Genomics

Year: 2015 PMID: 25909478 doi: 10.1186/s12864-015-1526-0 Citations: 24 Score: 12.11

Keywords hits: xyr1: 21, ace3: 1, cbh1: 2, cellulase: 49, reesei: 42

Abstract:
Trichoderma reesei is the main industrial source of cellulases and hemicellulases required for the hydrolysis of biomass to simple sugars, which can then be used in the production of biofuels and biorefineries. The highly productive strains in use today were generated by classical mutagenesis. As byproducts of this procedure, mutants were generated that turned out to be unable to produce cellulases. In order to identify the mutations responsible for this inability, we sequenced the genome of one of these strains, QM9136, and compared it to that of its progenitor T. reesei QM6a.

Summary:
Gene expression of the major cellulase cel7a, xylanase xyn2 and transcription factor xyr1 itself, induced by one hour exposure to 1.4 m M sophorose, are restored in QM9136:: GFP-XYR1 transformants. This finding is also nicely reflected in the phenotype of QM9136, which fails to grow on 65 m M D-xylose, has strongly reduced or no growth on lactose and cellulose, respectively, 31, and is deficient in the formation of cellulases, xylanases and mannanases 1. While the identification of a gene that is already known to be crucially involved in cellulase and hemicellulase formation, as the cause for the inability of cellulase formation of T. reesei QM9136, may be considered trivial, it nevertheless sheds new light into its molecular function: XYR1 belongs to the fungal-specific Zn2 Cys6-binuclear zinc cluster transcription factor family 32,33. This implies that this domain is essential for the activation of cellulase, xylanase and -mannanase gene expression. Growth of T. reesei QM6a and QM9136 on solid minimal agar medium supplemented with either D-glucose, cellobiose or lactose. Additional file 2: Table S1. Chromosomal location of mutated genes in QM9136. Additional file 3: Figure S2. c DNA alignment of native and mutated xyr1 loci. The truncated GFP-XYR1178 protein was detectable with an average intensity of about 4.5-times less compared to the full-length GFP-XYR1 construct.

Mentioned biomedical entities:
Genes: ACE2, ACE3, B, CRE1, ENS, HD, MA, SEB, XYR1, XlnR, Xyr1, cbh1, cellulase, cre1, ptrA, xyn1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, C.Escherichia coli strains, Cell, FB).The, GFP-XYR1 transformants, GFP-XYR1 transformants TRAL2, M1, Penicillium canescens, QM6a, QM6a GFP, QM9414::GFP-XYR1 transformants, RUT lineage, TA, TRAL2 clones 1, TRAL7 clones, TRAL7 clones 1, cell, cells, cellular, cellulases, cultures, cytosolic, hygromycin, ku7, line, live-cell, mycelium, small cellular, spore

Organisms: A. niger, mouse

Tissues: PDA hygromycin, ex-type, lens, tube

Pathways: N/A

Query Result 3/25

Trichoderma reesei XYR1 activates cellulase gene expression via interaction with the Mediator subunit TrGAL11 to recruit RNA polymerase II

Zheng, Fanglin; Cao, Yanli; Yang, Renfei; Wang, Lei; Lv, Xinxing; Zhang, Weixin; Meng, Xiangfeng; Liu, Weifeng; Butler, Geraldine

PLoS Genetics

Year: 2020 PMID: 32877410 doi: 10.1371/journal.pgen.1008979 Citations: 19 Score: 11.93

Keywords hits: xyr1: 97, ace3: , cbh1: 21, cellulase: 67, reesei: 3

Abstract:
The ascomycete Trichoderma reesei is a highly prolific cellulase producer. While XYR1 (Xylanase regulator 1) has been firmly established to be the master activator of cellulase gene expression in T. reesei, its precise transcriptional activation mechanism remains poorly understood. In the present study, TrGAL11, a component of the Mediator tail module, was identified as a putative interacting partner of XYR1. Deletion of Trgal11 markedly impaired the induced expression of most (hemi)cellulase genes, but not that of the major -glucosidase encoding genes. This differential involvement of TrGAL11 in the full induction of cellulase genes was reflected by the RNA polymerase II (Pol II) recruitment on their core promoters, indicating that TrGAL11 was required for the efficient transcriptional initiation of the majority of cellulase genes. In addition, we found that TrGAL11 recruitment to cellulase gene promoters largely occurred in an XYR1-dependent manner. Although xyr1 expression was significantly tuned down without TrGAL11, the binding of XYR1 to cellulase gene promoters did not entail TrGAL11. These results indicate that TrGAL11 represents a direct in vivo target of XYR1 and may play a critical role in contributing to Mediator and the following RNA Pol II recruitment to ensure the induced cellulase gene expression.

Summary:
Nevertheless, XYR1 occupancy on cellulase gene promoters was not affected in the absence of Tr GAL11, and elevated XYR1 expression was not able to rescue the defective cellulase gene expression in the Trgal11 deletion strain. Results Identification of a T. reesei GAL11/MED15 homolog that interacts with XYR1 To identify XYR1 coactivators and explore how they may facilitate XYR1-mediated transcriptional activation of cellulase genes in T. reesei, we focused on the Mediator complex whose tail module subunit GAL11/MED15 has been reported to serve as the target of various transcriptional activators 11, 12, 2, 39, 4. Were detected for bgl1 and bgl2 gene transcription between QM9414 and Trgal11 for the indicated time points after induction except bgl1 gene transcription induction for 12 h. To evaluate the relevant contribution of other putative tail module subunits to the induced cellulase gene expression, the identified Trmed3, Trmed5, and Trmed16 that have been shown to be nonessential in yeast 43, were individually deleted in T. reesei. Taken together, these data suggest that T. reesei Mediator may adopt a subtly different tail module organization from that reported for yeast to connect XYR1 to communicate the regulatory input to the Pol II enzyme. Tr GAL11 is recruited to cellulase gene promoters in an XYR1-dependent manner To ask whether the significantly decreased expression of the cbh/eg genes in the Trgal11 strain was caused by the down-regulated xyr1 transcripts, a recombinant strain that simultaneously overexpressed xyr1 under control of the tcu1 promoter in the Trgal11 strain was constructed. Significant differences were detected for XYR1 occupancy on cellulase gene promoters between QM9414 and Trgal11 mutant after Avicel induction for 6 h and 9 h. An overview of XYR1 occupancy over the cbh1 promoter analyzed by Ch IP after Avicel induction for 9 h. The analyzed regions are defined as in. Tr GAL11 contributes to RNA Pol II recruitment to cellulase gene core promoters As the Mediator tail module mainly functions to connect sequence-specific transcription factors to promote the formation of PIC, it was reasonable to believe that the absence of Tr GAL11 would compromise RNA Pol II recruitment to the core promoter upon cellulase gene activation. In contrast with cbh and eg genes, there likely exists an as yet to be identified transcriptional activator, which synergizes with XYR1 to achieve core Mediator recruitment to -glucosidase gene promoters by engaging interactions with subunit other than Tr GAL11.1.1371/journal. pgen.18979. g9 Fig 9 A schematic model of how Mediator is recruited by XYR1 to participate in activating cellulase gene expression in T. reesei. Once expressed upon induction, XYR1 binds to the upstream binding sites in cellulase gene promoters and recruits the Mediator complex by interacting with the tail module subunit Tr GAL11 to further facilitate the recruitment of the general transcription machinery including RNA Pol II to cbh and eg genes. Materials and methods Strains and cultivation condition Escherichia coli DH5 cells were used for plasmids construction and E. coli Origami BL21 cells were used as a host for the production of the recombinant proteins.

Mentioned biomedical entities:

Genes: FPA, GAL4, GCN4, GST, HSF1, MED14, MED16, MED2, MED5, NC, PIC, Rpb1, TATA, TFIIH, UPS, XYR1, actin, bgl1, bgl2, cbh1, cellulase, eg1, p53

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, E. coli, E. coli BL21, E. coli Origami BL21, F, F) Culture, GST, MED5, Yeast cells, cell lysates, cells, conidia, hygromycin B, malt extract, pGADT7-TrGAL11 FL, yeast cell, yeast cells, yeast transformant cells, yeast transformants

Organisms: E. coli, Liu lab, TURBO DNA-free kit, Tail, bovine serum albumin, human

Tissues: Tail

Pathways: N/A

Query Result 4/25

Inducer-free cellulase production system based on the constitutive expression of mutated XYR1 and ACE3 in the industrial fungus Trichoderma, breesei

Arai, Toshiharu; Ichinose, Sakurako; Shibata, Nozomu; Kakeshita, Hiroshi; Kodama, Hiroshi; Igarashi, Kazuaki; Takimura, Yasushi

Scientific Reports

Year: 2022 PMID: 36376415 doi: 10.1038/s41598-022-23815-4 Citations: 6 Score: 11.77

Keywords hits: xyr1: 136, ace3: 116, cbh1: 1, cellulase: 144, reesei: 25

Abstract:

Trichoderma, bc toggle=yesreesei is a widely used host for producing cellulase and hemicellulase cocktails for lignocellulosic biomass degradation. Here, we report a genetic modification strategy for industrial T., bc toggle=yesreesei that enables enzyme production using simple glucose without inducers, such as cellulose, lactose and sophorose. Previously, the mutated XYR1V821F or XYR1A824V was known to induce xylanase and cellulase using only glucose as a carbon source, but its enzyme composition was biased toward xylanases, and its performance was insufficient to degrade lignocellulose efficiently. Therefore, we examined combinations of mutated XYR1V821F and constitutively expressed CRT1, BGLR, VIB1, ACE2, or ACE3, known as cellulase regulators and essential factors for cellulase expression to the T., bc toggle=yesreesei E1AB1 strain that has been highly mutagenized for improving enzyme productivity and expressing a -glucosidase for high enzyme performance. The results showed that expression of ACE3 to the mutated XYR1V821F expressing strain promoted cellulase expression. Furthermore, co-expression of these two transcription factors also resulted in increased productivity, with enzyme productivity 1.5-fold higher than with the conventional single expression of mutated XYR1V821F. Additionally, that productivity was 5.5-fold higher compared to productivity with an enhanced single expression of ACE3. Moreover, although the DNA-binding domain of ACE3 had been considered essential for inducer-free cellulase production, we found that ACE3 with a partially truncated DNA-binding domain was more effective in cellulase production when co-expressed with a mutated XYR1V821F. This study demonstrates that co-expression of the two transcription factors, the mutated XYR1V821F or XYR1A824V and ACE3, resulted in optimized enzyme composition and increased productivity.

Summary:

The ace3 of E1 AB1-XA3 used in Fig. However, the productivity of the E1 AB1-XA3fl strain with FL-ACE3 was 14% lower than that of the E1 AB1-XA3 strain with PT-ACE3 (Fig. These results suggested that a certain level of total XYR1 or mutated XYR1, such as mutated XYR1 V821 F or XYR1 A824 V, was required for high cellulase production. These results also indicated that constitutive expression of the mutated XYR1 exhibiting a glucose-blind phenotype and ACE3 was necessary for high cellulase production in the absence of inducers. Additionally, the partial truncation of the DNA-binding domain of ACE3 and disruptions of ace1 and rce1 contributed to high protein productivity, and the genotype of the E1 AB1-XA3 strain was found to be the most effective combination in this study. Expression and activity analysis of cellulase and xylanase under single or co-expressed XYR1 V821 F and PT-ACE3 We analyzed the gene expression and activity of cellulase and xylanase in a single expression of XYR1 V821 F, a single expression of PT-ACE3, and co-expression of XYR1 V821 F and PT-ACE3. Owing to the effect of the XYR1 V821 F mutation, the transcription of major cellulases in the E1 AB1-X strain was approximately four orders of magnitude higher than that of the parent strain E1 AB1. Enhanced cellulase productivity using glucose as the sole carbon source was achieved by constitutive expression of mutated XYR1 V821 F or XYR1 A824 V and ACE3, especially ACE3, having a partially truncated DNA-binding domain. In this report, co-expression of XYR1 V821 F with cellulase regulation-related factors, CRT1, BGLR, VIB1, or ACE2, did not increase cellulase expression.

Mentioned biomedical entities:

Genes: ACE2, ACE3, ARE1, BXL1, CBH, CBH1, CBH2, CIP2, CRE1, CRT1, CRZ1, CTF1, Cellulase, EG1, Gal4, HD, KC, LAE1, MP, PAC1, RCE1, STR1, VEL1, VIB1, XYN1, XYN2, XYR1, ace1, ace2, act1, cbh1, cbh2, cellulase, cre1, pgk1, xyn1, xyn2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, CCR, Culture, E1AB1-A3, EG, FL-ACE3, PC-3-7, RL-P37, RUT-C3, Rut-C3, Spores, SuperScript IV VILO, TFs, b) SDS-PAGE, cell, cells, cellulases, gDNA, plasmid, wild-type

Organisms: bovine gamma globulin, wild-type XYR1

Tissues: tube

Pathways: N/A

Query Result 5/25

Enhancement of cellulase production in Trichoderma reesei RUT-C3 by comparative genomic screening

Liu, Pei; Lin, Aibo; Zhang, Guoxiu; Zhang, Jiajia; Chen, Yumeng; Shen, Tao; Zhao, Jian; Wei, Dongzhi; Wang, Wei

Microbial Cell Factories

Year: 2019 PMID: 31077201 doi: 10.1186/s12934-019-1131-z Citations: 16 Score: 10.96

Keywords hits: xyr1: 5, ace3: , cbh1: 2, cellulase: 82, reesei: 52

Abstract:
Cellulolytic enzymes produced by the filamentous fungus Trichoderma reesei are commonly used in biomass conversion. The high cost of cellulase is still a significant challenge to commercial biofuel production. Improving cellulase production in T. reesei for application in the cellulosic biorefinery setting is an urgent priority.

Summary:
Mutation of cre1 is not always the key step to increase total cellulase production in T. reesei because of the effect on the growth of T. reesei. Combining mutations of RUT-C3 and SS-II to produce more efficient cellulase-producing strains from T. reesei RUT-C3 The 57 genes with specific mutations in SS-II might be involved in cellulase hyper-production of SS-II. Of these five mutants, the 18642 strain displayed an 83.7% increase in FPase activity, which was the highest extracellular cellulase activity observed among the tested strains. The activity was approximately 13.26/nmol/min/mg using NADP+ as the coenzyme. However, the SS-II strain showed similar levels of FPase activity, and lower p NPCase activity compared with those of the RUT-C3 strain. One unit of enzymatic activity was defined as 1 mol of p-nitrophenol released from the substrate per min. The concentration of proteins in the supernatant was measured using a commercial protein assay kit based on the absorbance at 595 nm. Enzymatic hydrolysis of pretreated corn stover Hydrolysis of pretreated corn stover was performed as previously described 48.

Mentioned biomedical entities:
Genes: CRE1, Cellulase, GRD1, MA, PCS, bgl1, cbh1, cbh2, cellulase, cre1, hac1, sar1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Agrobacterium tumefaciens AGL1, KOs, RUT-C3, SS-II-cre196.All, corn stover, cre1SS-II, culture conditionsEscherichia coli, hygromycin, hygromycin B, lignin, spore

Organisms: Escherichia coli, biodetoxified corn stover

Tissues: Joint Genome

Pathways: N/A

Query Result 6/25

Identification of the CRE-1 Cellulolytic Regulon in Neurospora crassa, bCRE-1 Regulon

Sun, Jianping; Glass, N. Louise; Arkowitz, Robert Alan

PLoS ONE

Year: 2011 PMID: 21980519 doi: 10.1371/journal.pone.0025654 Citations: 91 Score: 10.88

Keywords hits: xyr1: 1, ace3: , cbh1: 3, cellulase: 3, reesei: 7

Abstract:

In filamentous ascomycete fungi, the utilization of alternate carbon sources is influenced by the zinc finger transcription factor CreA/CRE-1, which encodes a carbon catabolite repressor protein homologous to Mig1 from Saccharomyces cerevisiae. In Neurospora crassa, deletion of cre-1 results in increased secretion of amylase and -galactosidase.

Summary:

These data indicate that CCR in N. crassa was responsive to changes in cre-1 expression level, similar to findings with cre A in A. nidulans, brid=pone.25654-Strauss2 ref-type=bibr33.1.1371/journal. pone.25654. g2 Figure 2 Complementation and gene expression of cre-1. A) Pn-cre-1 and B) Pc-cre-1 strains were grown on different carbon sources at 3C for 24 hrs. Of these, 6 genes showed increased expression levels in the N. crassa cre-1 mutant under MM conditions, similar to H. jecorina, while 2 genes that showed increased expression level in H. jecorina cre1 mutant, but showed decreased expression levels in N. crassa. An additional 12-gene set showed increased expression level in the cre-1 mutant, were predicted to be secreted and had annotations associated with a role in plant cell wall degradation. Genes that showed increased expression in the cre-1 mutant under cellulolytic conditions that were positive by Ch IP-PCR included NCU734 and NCU2855, both of which are direct CRE1 targets in T. reesei, brid=pone.25654-Takashima1 ref-type=bibr25, 48. No cellulolytic genes were induced in the cre-1 mutant when grown in MM, consistent with the requirement for induction, as well as relief from CCR.1.1371/journal. pone.25654. g6 Figure 6 Relative expression levels of N. crassa genes predicted to be regulated by CRE-1. A) Genes with increased expression level in the cre-1 mutant as compared to WT under MM culture conditions expression levels of these genes are shown for cultures transferred to MM or Avicel for 4 hours.

Mentioned biomedical entities:

Genes: AlcR, CBH-1, CRE-1, CRE1, Cre1, CreA, GAPDH, LIC, Mig1, NCU721, NCU7326, NCU8785, PKA, PLA2, Snf1, Ysy6, act-1, actin, alcA, amylase, cbh-1, cbh1, ccg-1, cellulase, cre-1, creA, gh1-1, gh1-2, gh11-1, gh11-2, gh3-4, gh45-1, gh5-1, gh61-4, gh61-6, prnB, prnD, sPLA2, xlnR, xyn1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, F, F and, His+ GFP+ transformants, MM cultures, Penicillium canescens, Plasmid, amy, b strains, cbh-1, cell, cell wall material, cells, cellular, conidia, endoglucanase-2, his-3, mammalian cells, plant cell wall, plant cell wall polymers 1

Organisms: A. niger, A. oryzae, Am J Bot 3, Escherichia coli secY, Mouse IgG, Ryan F, S. cerevisiae YSY6, WT, his-3 locus

Tissues: tube

Pathways: N/A

Query Result 7/25

MAPkinases regulate secondary metabolism, sexual development and light dependent cellulase regulation in Trichoderma reesei

Schalamun, Miriam; Beier, Sabrina; Hinterdobler, Wolfgang; Wanko, Nicole; Schinnerl, Johann; Brecker, Lothar; Engl, Dorothea Elisa; Schmoll, Monika

Scientific Reports

Year: 2023 PMID: 36732590 doi: 10.1038/s41598-023-28938-w Citations: 11 Score: 10.87

Keywords hits: xyr1: 9, ace3: , cbh1: 15, cellulase: 48, reesei: 4

Abstract:
The filamentous fungus Trichoderma reesei is a prolific producer of plant cell wall degrading enzymes, which are regulated in response to diverse environmental signals for optimal adaptation, but also produces a wide array of secondary metabolites. Available carbon source and light are the strongest cues currently known to impact secreted enzyme levels and an interplay with regulation of secondary metabolism became increasingly obvious in recent years. While cellulase regulation is already known to be modulated by different mitogen activated protein kinase (MAPK) pathways, the relevance of the light signal, which is transmitted by this pathway in other fungi as well, is still unknown in T. reesei as are interconnections to secondary metabolism and chemical communication under mating conditions. Here we show that MAPkinases differentially influence cellulase regulation in light and darkness and that the Hog1 homologue TMK3, but not TMK1 or TMK2 are required for the chemotropic response to glucose in T. reesei. Additionally, MAPkinases regulate production of specific secondary metabolites including trichodimerol and bisorbibutenolid, a bioactive compound with cytostatic effect on cancer cells and deterrent effect on larvae, under conditions facilitating mating, which reflects a defect in chemical communication. Strains lacking either of the MAPkinases become female sterile, indicating the conservation of the role of MAPkinases in sexual fertility also in T. reesei. In summary, our findings substantiate the previously detected interconnection of cellulase regulation with regulation of secondary metabolism as well as the involvement of MAPkinases in light dependent gene regulation of cellulase and secondary metabolite genes in fungi., b

Summary:
Specific cellulase activity upon growth on 1% cellulose in darkness. Transcript levels of cre1 upon growth on 1% cellulose in constant darkness and constant light. Since also the GPCRs CSG1 and CSG2 are required for chemotropic reactions to glucose84, the signaling pathway triggering this reaction in T. reesei might not be exclusively channeled through the G-protein pathway but may be subject to biased signaling85. MAPkinases regulate cellulase transcription and secreted activity differentially in light and darkness An involvement of T. reesei MAPkinases in cellulase regulation was shown previously7476. Therefore, we asked whether the regulatory function of the MAPkinases might be connected to the role of VEL1 by testing transcript abundance of vel1 in deletion strains of tmk1, tmk2 and tmk3. Indeed, we found a light dependent regulation of vel1 in all MAPkinase mutants, with differential impacts either in constant light or in constant darkness. We therefore checked available transcriptome data from comparable conditions for indications if such a feedback might exist5,112, but since we did not find significant regulation of tmk1, tmk2 or tmk3 in these data, we conclude that this is not the case. Interestingly, in N. crassa the OS pathway, corresponding to the Hog1-pathway in yeast and comprising a homologue of TMK3 has no significant influence on cellulase production13, which is in contrast to our results. In summary, our data obtained with experiments under controlled light conditions clearly show a light dependent regulatory function of all three MAPkinases on cellulase gene regulation and secreted cellulase activity, which is jeopardized by random light pulses. The GPCR CSG1, which is essential for the chemotropic response of T. reesei to glucose84, was shown to be required for posttranscriptional regulation of cellulase gene expression78. All three mutants were confirmed to only have a single integration of the deletion cassette by copy number determination12. Crossing and selection for fully fertile progeny for assessment of sexual development All crosses for the analysis of sexual development were performed on 6 mm 2% MEX agar plates at 22 C and 12 h lightdark cycles as previously described116.

Mentioned biomedical entities:

Genes: B, CSG2, CWI, Fus3, Gpa2, MAPK, MAT1-1, MAT1-2, Slt2, Tmk1, Tmk2, VEL1, XYR1, cbh1, cellulase, cre1, env1, peroxidase, pks4, sar1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: F, F) Transcript, QM6a, S., TMK1, Tmk1, aflatoxin, cancer cell lines, cell wall, cells, cellular, hygromycin, liquid, liquid cultures, malt extract, photoreceptors, plant cell wall, solid, spore

Organisms: A. flavus, CBS999.97 MAT1-2, HR ESI MS

Tissues: Ascospore, capillary, lateral, layer

Pathways: N/A

Query Result 8/25

Characterization of a novel sugar transporter involved in sugarcane bagasse degradation in Trichoderma reesei

Nogueira, Karoline M. V.; de Paula, Renato Graciano; Antonito, Amanda Cristina Campos; dos Reis, Thaila F.; Carraro, Cludia Batista; Silva, Alinne Costa; Almeida, Fausto; Rechia, Carem Gledes Vargas; Goldman, Gustavo H.; Silva, Roberto N.

Biotechnology for Biofuels

Year: 2018 PMID: 29619080 doi: 10.1186/s13068-018-1084-1 Citations: 30 Score: 10.72

Keywords hits: xyr1: 1, ace3: 1, cbh1: , cellulase: 19, reesei: 49

Abstract:

Trichoderma reesei is a saprophytic fungus implicated in the degradation of polysaccharides present in the cell wall of plants. T. reesei has been recognized as the most important industrial fungus that secretes and produces cellulase enzymes that are employed in the production of second generation bioethanol. A few of the molecular mechanisms involved in the process of biomass deconstruction by T. reesei; in particular, the effect of sugar transporters and induction of xylanases and cellulases expression are yet to be known.

Summary:

Transformants were selected for tryptophan, leucine, and uridine prototrophy, and the yeast candidates were confirmed by PCR using the primers 69957-F and 69957-R. A list of all the primers used in this study can be accessed in Additional file 5. Growth of S. cerevisiae strain on solid medium Saccharomyces cerevisiae strains were inoculated in 5 m L YNB medium supplemented with 2% maltose for 48 h at 3 C, 2 rpm. 2 Effect of Tr69957 transporter on growth of S. cerevisiae in the presence of cellobiose, mannose, and xylose. Analysis of the biomass and sugar consumption of parental and mutant strains under culture in MAM supplemented with mannose, cellobiose, and xylose, during growth for , 6, 24, and 48 h in the presence of 1% mannose, cellobiose, or xylose. Analysis of the expression of cel6a, xyn1, Tr69957, cel7a, cel1a, cel3a, and xyn2 during cultivation of the mutant and parental strains grown in MAM supplemented with cellobiose, mannose, xylose, or SCB as the sole carbon source. 7, in the presence of mannose, the expression of -glucosidase is higher than that of other cellulases, indicating that mannose present in the environment might signal the expression of these enzymes, which act in the conversion of cellobiose to sophorose 55, thereby increasing the cellulase expression during the late signaling of Tr69957. In this study, we characterized a cellobiose, mannose, and xylose transporter involved in SCB degradation in T. reesei.

Mentioned biomedical entities:

Genes: ACE2, BGL1, CRE1, GLUT3, MFS, STP1, Sar1, XKS1, XR, cellulase, gh1-1, sar, xyn1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: 24-wells, CBC, Cells, EC, Penicillium, QM6a, TF, TFs, TU, Yeast cells, b, b id=Par44, b, cell, cell 7, cell suspensions, cells, cellulases, colonies, lines, plasmid, solid, solid YNB, solid yeast nitrogen base, yeast S., yeast S. cerevisiae, yeast cell surface, yeast cells

Organisms: A. niger, H Minus, S. cerevisiae, human glucose, humans, pGH1 plasmid

Tissues: column, lens

Pathways: N/A

Query Result 9/25

Ras GTPases Modulate Morphogenesis, Sporulation and Cellulase Gene Expression in the Cellulolytic Fungus Trichoderma reesei, bRas Modulate Morphogenesis and Cellulase Formation

Zhang, Jiwei; Zhang, Yanmei; Zhong, Yaohua; Qu, Yinbo; Wang, Tianhong; Xue, Chaoyang

PLoS ONE

Year: 2012 PMID: 23152805 doi: 10.1371/journal.pone.0048786 Citations: 16 Score: 10.62

Keywords hits: xyr1: 47, ace3: , cbh1: 25, cellulase: 69, reesei: 5

Abstract:

The model cellulolytic fungus Trichoderma reesei (teleomorph Hypocrea jecorina) is capable of responding to environmental cues to compete for nutrients in its natural saprophytic habitat despite its genome encodes fewer degradative enzymes. Efficient signalling pathways in perception and interpretation of environmental signals are indispensable in this process. Ras GTPases represent a kind of critical signal proteins involved in signal transduction and regulation of gene expression. In T. reesei the genome contains two Ras subfamily small GTPases TrRas1 and TrRas2 homologous to Ras1 and Ras2 from S. cerevisiae, but their functions remain unknown.

Summary:

Interestingly, Tr Ras2 could not produce clear zone around the colony on cellulose plate as compared with the parental strain and Tr Ras1, which suggests that Tr Ras2 may be involved in regulation of cellulase production.1.1371/journal. pone.48786. g3 Figure 3 Growth of TU-6, Tr Ras1 and Tr Ras2 on different carbon sources. Strains were grown on plates containing minimal medium supplemented with 1% glucose, glycerol, lactose or cellulose at 3C for 4 days. Tr Ras1 is essential for polarized apical growth, branching and sporulation When cultured on complete medium PDA, the Tr Ras1 mutant formed dramatically decreased colonies with dense mycelia and no conidiospores which was similar to that observed on minimal medium agar plate in. Scale bars are shown in the figure. The effects of Tr Ras2 on morphological phenotypes were analyzed by culturing the parental strain TU-6-Z and relevant mutants on PDA plates or liquid minimal medium with glucose as the carbon source. These data suggest that Tr Ras2 may sense the signals except cellulose and act upstream of the transcription regulators to modulate cellulase gene expression. The expression of the cellulase gene transcription factors is influenced by Tr Ras2 It is believed that the expression of cellulolytic gene is regulated by the cellulase transcription factors in T. reesei, brid=pone.48786-Stricker2 ref-type=bibr13, 14, 17, 4. Additionally, we also found Tr Ras2 has a negative effect on transcript level of ace1 but a positive influence on cre1 transcription only in the presence of cellulose, while only has a small effect on transcript abundance of ace2.1.1371/journal. pone.48786. g7 Figure 7 Analysis of the influence of Tr Ras2 on expression of the transcription factors encoding genes. Relative expression of xyr1, ace1, cre1 and ace2 from the parental strain TU-6-Z, the Tr Ras2 strain and the PAnigpd A-Tr Ras2 G16 V strain were shown. However, no alteration in the c AMP level on cellulose or glucose is detected in the mutant with a deletion for Tr Ras2, indicating that influence of Tr Ras2 on cellulase gene transcription is independent of c AMP signalling pathway. Xyr1 has been demonstrated as a central transcriptional regulator that controls xylanolytic as well as cellulolytic enzyme genes expression in T. reesei, brid=pone.48786-Stricker2 ref-type=bibr13.

Mentioned biomedical entities:

Genes: Ace1, Ace2, Actin, B, Cdc42, Cellulase, Cre1, GNA3, MAP, MAPK, PKAC1, Ras, Ras1, Ras2, RasA, RasB, RasC, Rho, Xyr1, ace1, ace2, actin, cAMP, cbh1, cbh2, cellulase, cre1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: PAnigpdA-TrRas2G16V strains, PAnigpdA-TrRas2G16V transformant, S., S. cerevisiae Ras1, Solid lines, TU-6, TU-6-Z, TU-6-Z transformant, U., U. maydis, b strains, b, black lines, cbh1-TrRas1 transformant, cell, cells, cellular, cellulose, cellulose-cellulase, cerevisiae, colonies, conidia, hyphae, hyphal cell, hyphal cells, lines, plant cell wall, pseudohyphal, solid agar, strains TU-6, strains TU-6-Z, wild-type, yeast cells

Organisms: A., C., C. albicans, C. neoformans, H., S. cerevisiae Ras1, human, human pathogens, human tumors

Tissues: PDA plate, U. maydis, peripheral

Pathways: N/A

Query Result 10/25

Tailoring the expression of Xyr1 leads to efficient production of lignocellulolytic enzymes in Trichoderma reesei for improved saccharification of corncob residues

Shen, Linjing; Yan, Aiqin; Wang, Yifan; Wang, Yubo; Liu, Hong; Zhong, Yaohua

Biotechnology for Biofuels and Bioproducts

Year: 2022 PMID: 36528622 doi: 10.1186/s13068-022-02240-9 Citations: 3 Score: 10.45

Keywords hits: xyr1: 124, ace3: 2, cbh1: 28, cellulase: 96, reesei: 5

Abstract:
The filamentous fungus Trichoderma reesei is extensively used for the industrial-scale cellulase production. It has been well known that the transcription factor Xyr1 plays an important role in the regulatory network controlling cellulase gene expression. However, the role of Xyr1 in the regulation of cellulase expression has not been comprehensively elucidated, which hinders further improvement of lignocellulolytic enzyme production.

Summary:
Furthermore, the overexpression of xyr1 with Pegl2 also remarkably improved the levels of accessory proteins involved in lignocellulose degradation and thus promoted the saccharification efficiency toward acid-pretreated corncob residues. Results Overexpression of the xyr1 gene using the promoters with different strengths in T. reesei To systematically investigate the effect of differential expression of xyr1 on cellulase production in T. reesei, the xyr1 overexpression cassettes driven by the cellulose-inducible promoters Pegl2, Pcbh1, and the strong constitutive promoter Pcdna1 were constructed and precisely integrated by homologous recombination into the hph locus of T. reesei QEB4, respectively. Genes encoding cellulases and transcription factors were analyzed in the xyr1 overexpression strains with T. reesei QEB4 as the control. These results indicate that the Pegl2-driven overexpression of xyr1 can optimize the lignocellulolytic enzyme system of T. reesei by improving the expression of those accessory proteins, which is beneficial for efficient degradation of lignocellulosic biomass. Fig. Reesei QEB4 T. A Compared to the strong xyr1 overexpression with the cbh1/cdna1 promoter, the moderate xyr1 overexpression with the egl2 promoter can result in more open chromatin and stronger expression of cellulase gene, thus significantly enhancing cellulase production.

Mentioned biomedical entities:
Genes: Ace1, Ace2, Ace3, B, BGL1, BglR, CBH, Cip1, Cip2, Cre1, Crz1, EG2, ER, FPA, Pac1, Ppdc, Rce1, Sec16, Swo1, Vib1, Xyr1, ace1, actin, bgl1, cbh1, cbh2, cellulase, cip1, cip2, cre1, histone, hrd1, pdi1, ptrA, swo1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: ACR, C18 Sep-Pak, E, EC, EG, clone, colonies, layer

Organisms: bovine serum albumin

Tissues: N/A

Pathways: N/A

Query Result 11/25

Weighted Gene Co-expression Network Analysis Identifies Critical Genes for the Production of Cellulase and Xylanase in Penicillium oxalicum

Li, Cheng-Xi; Zhao, Shuai; Luo, Xue-Mei; Feng, Jia-Xun

Frontiers in Microbiology

Year: 2020 PMID: 32292397 doi: 10.3389/fmicb.2020.00520 Citations: 13 Score: 10.18

Keywords hits: xyr1: 2, ace3: , cbh1: 1, cellulase: 81, reesei: 4

Abstract:

Genes involved in cellular processes undergo environment-dependent co-regulation, but the co-expression patterns of fungal cellulase and xylanase-encoding genes remain unclear. Here, we identified two novel carbon sources, methylcellulose and 2-hydroxyethyl cellulose, which efficiently induced the secretion of cellulases and xylanases in Penicillium oxalicum. Comparative transcriptomic analyses identified carbon source-specific transcriptional patterns, mainly including major cellulase and xylanase-encoding genes, genes involved in glycolysis/gluconeogenesis and the tricarboxylic acid cycle, and genes encoding transcription factors, transporters and G protein-coupled receptors. Moreover, the weighted correlation network analysis of time-course transcriptomes, generated 17 highly connected modules. Module MEivory, comprising 12 members, included major cellulase and xylanase-encoding genes, genes encoding the key regulators PoxClrB and PoxXlnR, and a cellodextrin transporter POX651/CdtC, which were tightly correlated with the filter-paper cellulase, carboxymethylcellulase and xylanase activities in P. oxalicum. An expression kinetic analysis indicated that members in MEivory were activated integrally by carbon sources, but their expressional levels were carbon source- and/or induction duration-dependent. Three uncharacterized regulatory genes in MEivory were identified, which regulate the production of cellulases and xylanases in P. oxalicum. These findings provide insights into the mechanisms associated with the synthesis and secretion of fungal cellulases and xylanases, and a guide for P. oxalicum application in biotechnology.

Summary:

DEGs, differentially expressed genes AV, Avicel MC, methyl cellulose HEC, 2-hydroxyethyl cellulose GLU, glucose NC, no carbon source. As expected, 1628 of the 35 cellulase and endo-xylanase genes in the genome of HP7-1 showed significant alterations at the transcriptional level in response to AV, WB, MC, and HEC for 4 to 48 h, compared with that on NC, including the known key cellulase and xylanase-encoding genes, such as POX5587/Cel7 A-2, POX557/Cel45 A, POX6147/Cel5 A, POX7535/Cel12 A, POX6835/Bgl1, POX63/Xyn1 A, and POX6783/Xyn11 A most of them were up-regulated. Conversely, during growth on GLU, 917 of 35 cellulase and xylanase-encoding genes showed alteration compared with that on NC, most of them down-regulated. oxalicum on different carbon sources compared with NC. Their transcriptional levels varied in response to the different carbon sources and the duration of induction, for instance, 44 and 165 DEGs on MC and WB at 4 h compared with that on NC, respectively, and 34 and 51 at 12 h. Screening for the top ten upregulated/downregulated TF genes for each comparison detected a total of 163, including several known regulatory genes, i. e., POX1167/Pox Cxr A, POX442/Pox Cxr B, POX196/Pox Clr B, POX2484, POX389/Pox Amy R, POX8292/Pox MBF1, and POX6534/Brl A. oxalicum on different carbon sources compared with NC. However, these annotated genes were not previously known to be associated with cellulase and/or xylanase production in P. oxalicum. The remaining five modules MEmidnightblue, MEorange, MEturquoise, MEgrey, and MElightcyan1 contained a few unimportant cellulase and XYN genes, and their members accounted for approximately 8% of all genes annotated in the genome of HP7-1, which should not therefore, be co-expressed with the major cellulase and XYN genes. In order to understand the significance of the identified modules toward cellulase and xylanase production of P. oxalicum, their correlation coefficients were calculated. To facilitate further study, these 12 DEGs were named as basic co-expression genes under biomass induction. In addition, all 12 DEGs increased their transcriptional levels to various degrees under the induction of AV, MC, HEC and WB, with log2 fold changes from 1.1 to 1.3, except POX1833 on WB at 4 h. Conversely, most of them showed a significant reduction on GLU. In addition to these 12 DEGs that were induced commonly by biomass, the expression levels of several specific DEGs showed significant alterations that were dependent on the duration of induction and/or different substrates.

Mentioned biomedical entities:

Genes: ADA2, AbaA, ArpA, BGLs, CBH, Cellulase, Clp protease, ClpP, CreA, FlbB, FlbD, GLU, GskA, HEC, LaeA, Mbp1, MreA, NC, NimX, TmpA, XlnR, Xyr1, amylase, cbh, cbh1, cellulase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: AC, Culture, Culture ConditionsThe fungal, Fusarium oxysporum, GLU, HEC, POX196/ClrB, Penicillium, TF, TFs, Talaromyces cellulolyticus Y-94, cell, cell wall-degrading enzymes, cells, cellulose, plant cell walls, spore, starch

Organisms: N/A

Tissues: brown, modular TFs, tissue

Pathways: N/A

Query Result 12/25

The Different Roles of Penicillium oxalicum LaeA in the Production of Extracellular Cellulase and -xylosidase

Li, Yanan; Zheng, Xiaoju; Zhang, Xiujun; Bao, Longfei; Zhu, Yingying; Qu, Yinbo; Zhao, Jian; Qin, Yuqi

Frontiers in Microbiology

Year: 2016 PMID: 28066400 doi: 10.3389/fmicb.2016.02091 Citations: 21 Score: 10.15

Keywords hits: xyr1: 12, ace3: , cbh1: 3, cellulase: 51, reesei: 13

Abstract:

Cellulolytic enzyme hydrolysis of lignocellulose biomass to release fermentable sugars is one of the key steps in biofuel refining. Gene expression of fungal cellulolytic enzymes is tightly controlled at the transcriptional level. Key transcription factors such as activator ClrB/CLR2 and XlnR/XYR1, as well as repressor CreA/CRE1 play crucial roles in this process. The putative protein methyltransferase LaeA/LAE1 has also been reported to regulate the gene expression of the cellulolytic enzyme. The formation and gene expression of the cellulolytic enzyme was compared among Penicillium oxalicum wild type (WT) and seven mutants, including laeA (deletion of laeA), OEclrB (clrB overexpression), OEclrBlaeA (clrB overexpression with deletion of laeA), OExlnR (xlnR overexpression), OExlnRlaeA (xlnR overexpression with deletion of laeA), creA (deletion of creA), and creAlaeA (double deletion of creA and laeA). Results revealed that LaeA extensively affected the expression of glycoside hydrolase genes. The expression of genes that encoded the top 1 glycoside hydrolases assayed in secretome was remarkably downregulated especially in later phases of prolonged batch cultures by the deletion of laeA. Cellulase synthesis of four mutants laeA, OEclrBlaeA, OExlnRlaeA, and creAlaeA was repressed remarkably compared with their parent strains WT, OEclrB, OExlnR, and creA, respectively. The overexpression of clrB or xlnR could not rescue the impairment of cellulolytic enzyme gene expression and cellulase synthesis when LaeA was absent, suggesting that LaeA was necessary for the expression of cellulolytic enzyme gene activated by ClrB or XlnR. In contrast to LaeA positive roles in regulating prominent cellulase and hemicellulase, the extracellular -xylosidase formation was negatively regulated by LaeA. The extracellular -xylosidase activities improved over 5-fold in the OExlnRlaeA mutant compared with that of WT, and the expression of prominent -xylosidase gene xyl3A was activated remarkably. The cumulative effect of LaeA and transcription factor XlnR has potential applications in the production of more -xylosidase.

Summary:

Several transcription factors such as Cre A, Clr B, and Xln R have been found to play important roles in regulating cellulolytic enzyme gene expression. For instance, the overexpression of clr B, combined with both deletion of cre A and an intracellular -glucanse gene bgl2, remarkably activated the cellulase gene expression and improved filter paper activity by over 2-fold. In this study, we investigated the roles of P. oxalicum Lae A in regulating the expression of cellulase and -xylosidase gene, especially when transcription activator gene was overexpressed or transcription repressor gene was deleted. The primers that were used to examine gene expression levels included act-F/act-R, cel7 A-F/cel7 A-R, cel7 B-F/cel7 B-R, cel3 A-F/cel3 A-R, amy15 A-F/amy15 A-R, xln1 A-F/xln1 A-R, xyl3 A-F/xyl3 A-R, lae A-F/lae A-R, cre A-F/cre A-R, clr B-F/clr B-R, and xln R-F/xln R-R. FPA of WT and lae A. Then, the transcription levels of genes cre A, clr B, and xln R were quantified in WT and the mutants grown from 12, 24, 36, 48, and 6 h.

Mentioned biomedical entities:

Genes: ACE1, AD, BD, Bok, CRE1, Cel61A, Cellulase, CreA, FPA, GAL4, LAE1, Lae1, LaeA, Met, Rel, VelB, XYR1, XlnR, Xyl3A, Xyr1, actin, bgl2, cbh1, cellulase, creA, gpdA, histone, lacZ, laeA, ptrA, xlnR, xyn1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: 4-day-old cultures, C, Conidial, MnSO4H2O, Penicillium, Penicillium chrysogenum, SYBR Premix Ex Taq, ade2, cell, cells, cellulose, clones, conidial, conidium, dark-green colonies, protoplast, wheat bran

Organisms: A. oryzae, Clontech, WT, WT colony

Tissues: N/A

Pathways: N/A

Query Result 13/25

Inducible promoters and functional genomic approaches for the genetic engineering of filamentous fungi

Kluge, Janina; Terfehr, Dominik; Kck, Ulrich

Applied Microbiology and Biotechnology

Year: 2018 PMID: 29860590 doi: 10.1007/s00253-018-9115-1 Citations: 40 Score: 10.11

Keywords hits: xyr1: 2, ace3: , cbh1: 4, cellulase: 8, reesei: 18

Abstract:

In industry, filamentous fungi have a prominent position as producers of economically relevant primary or secondary metabolites. Particularly, the advent of genetic engineering of filamentous fungi has led to a growing number of molecular tools to adopt filamentous fungi for biotechnical applications. Here, we summarize recent developments in fungal biology, where fungal host systems were genetically manipulated for optimal industrial applications. Firstly, available inducible promoter systems depending on carbon sources are mentioned together with various adaptations of the Tet-Off and Tet-On systems for use in different industrial fungal host systems. Subsequently, we summarize representative examples, where diverse expression systems were used for the production of heterologous products, including proteins from mammalian systems. In addition, the progressing usage of genomics and functional genomics data for strain improvement strategies are addressed, for the identification of biosynthesis genes and their related metabolic pathways. Functional genomic data are further used to decipher genomic differences between wild-type and high-production strains, in order to optimize endogenous metabolic pathways that lead to the synthesis of pharmaceutically relevant end products. Lastly, we discuss how molecular data sets can be used to modify products for optimized applications.

Summary:

This is in part due to the fact that the expression machinery is more complex in filamentous fungi than in bacteria and yeasts. As an advantage, filamentous fungi tend to have post-translational modification systems that allow the production of functional mammalian proteins, which are difficult to perform in bacterial and yeast expression systems in many cases. In a recent study, molecular tools, such as homologous recombination and RNA interference systems that are suitable for the genetic manipulation of filamentous fungi were reviewed. In addition, a leakiness of the promoter, which results in a minimal constitutive activity, leads to uncontrolled gene expression. Few of them that were used in recent biotechnical applications will be summed up in this review. Inducible promoters depending on carbon sources The gla A promoter of the glucoamylase A gene from Aspergillus niger was one of the first inducible systems, which was commonly used in filamentous fungi. The inducible overexpression of TF22 results in activation of the three cluster genes indicating that this transcription factor directs the biosynthesis of trichosetin. Optimization of the Tet-Off system in A. niger was achieved by using the present plasmid for the Tet-On system and the reverse transactivator rt TA2 S-M2 against the t TAS2 transactivator was exchanged, which is optimized for mammals, in order to construct a functional Tet-Off cassette (Fig. Although the achieved yields cannot compete with established production hosts like P. pastoris, which is able to produce up to 1-fold higher yields, N. crassa seems to be a promising new host system for the production and secretion of heterologous proteins. Strain improvement using genomics and functional genomics The strain improvement programs of A. chrysogenum, P. chrysogenum, and T. reesei are suitable examples for the efficiency of classical strain improvement by random mutagenesis.

Mentioned biomedical entities:

Genes: Atg1, Bok, CHY, Cre1, PrtT, TetA, TetR, VP16, Xyr1, alcA, bphA, cbh1, cellulase, cre1, gla1, glaA, gpdA, niiA, qa-2, tTA, tetA, thiA, vvd, xyl1, xyn1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Camembert-type cheese, E. coli, Hansenula polymorpha, Penicillium camemberti, Penicillium chrysogenum, Penicillium species, Pichia pastoris, RUT-C3, Rut-C3, Tada, Tet-Off, Tet-On, Trichoderma, basal, cell, cellular, conidia, cultures, dendritic cells, gene-conditional, hygromycin-resistant transformants, pastoris, stem

Organisms: A., A. chrysogenum, A. niger, A. oryzae, Blue light, Blue light-inducible, Escherichia coli, bovine chymosin, herpes simplex virus, human, human hormone peptide, human lysozyme, mice, murine, pulmonary

Tissues: Tet construct, Tet system, Tet system).

Pathways: N/A

Query Result 14/25

Diversity of Cellulase-Producing Filamentous Fungi From Tibet and Transcriptomic Analysis of a Superior Cellulase Producer Trichoderma harzianum LZ117

Li, Jia-Xiang; Zhang, Fei; Jiang, Dan-Dan; Li, Jun; Wang, Feng-Lou; Zhang, Zhang; Wang, Wei; Zhao, Xin-Qing

Frontiers in Microbiology

Year: 2020 PMID: 32760377 doi: 10.3389/fmicb.2020.01617 Citations: 9 Score: 10.06

Keywords hits: xyr1: 12, ace3: 2, cbh1: 4, cellulase: 76, reesei: 21

Abstract:

Filamentous fungi are widely used for producing cellulolytic enzymes to degrade lignocellulosic biomass. Microbial resources from Tibet have received great attention due to the unique geographic and climatic conditions in the Qinghai-Tibet Plateau. However, studies on cellulase producing fungal strains originated from Tibet remain very limited, and so far no studies have been focused on regulation of cellulase production of the specific strains thereof. Here, filamentous fungal strains were isolated from soil, plant, and other environments in Tibet, and cellulase-producing strains were further investigated. A total of 88 filamentous fungal strains were identified, and screening of cellulase-producing fungi revealed that 16 strains affiliated with the genera Penicillium, Trichoderma, Aspergillus, and Talaromyces exhibited varying cellulolytic activities. Among these strains, T. harzianum isolate LZ117 is the most potent producer. Comparative transcriptome analysis using T. harzianum LZ117 and the control strain T. harzianum K223452 cultured on cellulose indicated an intensive modulation of gene transcription related to protein synthesis and quality control. Furthermore, transcription of xyr1 which encodes the global transcriptional activator for cellulase expression was significantly up-regulated. Transcription of cre1 and other predicted repressors controlling cellulase gene expression was decreased in T. harzianum LZ117, which may contribute to enhancing formation of primary cellulases. To our knowledge, this is the first report that the transcription landscape at the early enzyme production stage of T. harzianum was comprehensively described, and detailed analysis on modulation of transporters, regulatory proteins as well as protein synthesis and processing was presented. Our study contributes to increasing the catalog of publicly available transcriptome data from T. harzianum, and provides useful clues for unraveling the biotechnological potential of this species for lignocellulosic biorefinery.

Summary:

To further verify the reliability of transcriptome data, gene transcription analysis of T. harzianum LZ117 over T. harzianum K223452 at 24 h was performed by RT-q PCR for several genes, including the transcription factor genes xyr1 and cre1, as well as glycosyl hydrolase genes cbh1, egl1, egl2, and bgl1. Comparative transcriptome analysis showed that there were 116 DEGs between T. harzianum LZ117 and K223452 at 24 h of cellulose induction, of which 375 genes were up-regulated and 785 genes were down-regulated. Gene ontology functional enrichment analysis of significantly up-regulated DEGs showed that not only genes involved in cellular process like protein synthesis, processing and degradation, but also metabolic activities of the cell, including subcategories of purine nucleotide metabolism, carbohydrate biosynthesis, and energy metabolism presented significant enrichment. Energy metabolism is important for protein synthesis, and we assume that up-regulation of the related genes may benefit cellulase production in LZ117. Other novel sugar transporters that are crucial for cellulase induction signaling in T. harzianum will be further investigated in future studies. DEGs Related to Protein Synthesis, Sorting and Quality Control The functional classification analysis for DEGs indicated that the transcriptional levels of 78.6% of the DEGs involved in transcription and protein synthesis in T. harzianum LZ117 strains were up-regulated as compared to T. harzianum K223452. Genes identified in this study can be further analyzed and used to improve cellulase production in other T. harzianum strains as well as T. reesei strains.

Mentioned biomedical entities:

Genes: Ace1, Ace3, CBH, CBH1, Cdt1, Cdt2, Cellulase, Cre1, Crt1, Ctf1, FPA, FZ, GH, GH18, GHs, GPI, Hsp3, MA, Rce1, SRA, Vib1, Xyr1, ace1, axe1, bgl1, cbh1, cdt1, cdt2, cellulase, cre1, crt1, xyn1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, Culture, Fusarium strains, GH3, MEA, Penicillium, TFs, Trichoderma, Trichoderma strains, barley wine, beta-mannosidase7.73721MannosidaseGH76, cell, cell wall-degrading enzymes, cellular, chaperones, malt extract, mushroom

Organisms: Kato

Tissues: Joint Genome, Tissue, tube

Pathways: N/A

Query Result 15/25

Exploring the Synergy between Cellobiose Dehydrogenase from Phanerochaete chrysosporium and Cellulase from Trichoderma reesei

Wang, Min; Lu, Xuefeng

Frontiers in Microbiology

Year: 2016 PMID: 27199949 doi: 10.3389/fmicb.2016.00620 Citations: 12 Score: 10.03

Keywords hits: xyr1: 1, ace3: , cbh1: 6, cellulase: 82, reesei: 37

Abstract:

Recent demands for the production of lignocellulose biofuels boosted research on cellulase. Hydrolysis efficiency and production cost of cellulase are two bottlenecks in biomass to biofuels process. The Trichoderma cellulase mixture is one of the most commonly used enzymes for cellulosic hydrolysis. During hydrolytic process cellobiose accumulation causes feedback inhibition against most cellobiohydrolases and endoglucanases. In this study, we demonstrated the synergism effects between cellobiose dehydrogenase (CDH) and cellulase both in vitro and in vivo. The CDH from Phanerochaete chrysosporium was heterologously expressed in Pichia pastoris. Supplementation of the purified CDH in Trichoderma cellulase increased the cellulase activities. Especially -glucosidase activity was increased by 31% varying at different time points. On the other hand, the cdh gene was heterologously expressed in Trichoderma reesei to explore the synergism between CDH and cellulases in vivo. The analyses of gene expression and enzymatic profiles of filter paper activity, carboxymethylcellulase (CMCase) and -glucosidase show the increased cellulase activity and the enhanced cellulase production in the cdh-expressing strains. The results elucidate a possible mechanism for diminishing the cellobiose inhibition of cellulase by CDH. These findings provide a novel perspective to make more economic enzyme cocktails for commercial application or explore alternative strategies for generating cellulase-producing strains with higher efficiency.

Summary:

Accumulation of cellobiose causes feedback inhibition against most cellobiohydrolases and endoglucanases, and thus it becomes important to supply enough -glucosidases to hydrolyze cellobiose to glucose in order to eliminate such inhibition. In T. reesei cellulase system, the two most abundant proteins are cellobiohydrolases CBH I and CBH II, which are known to account for 7 to 8% of the total cellulases. To determine whether the increased cellulase activity was also accompanied with the changes in cellulase expression level compared with parental QM9414, we analyzed the extracellular proteins of the cdh-expressing and the native QM9414 strains by SDS-PAGE, and the secreted proteins were quantified by Bradford assay. Relatively high levels of expression of these -glucosidase genes in cdh-expressing strain implied that it might be caused by cellobiose degradation. In Vivo Synergistic Effects between CDH and Cellulase with Cellulose as the Carbon Source As the results showed above, the synergism between CDH and cellulase was observed in vivo in fermentation process with lactose as the soluble carbon source. At 96 h, the intracellular protein of cdh:: QM9414 decreased about 5% compared to the parental QM9414 strain. The extracellular protein content was determined by Bradford assay and the expression profile was analyzed on SDS-PAGE. In addition, although the cdh was controlled under the cbh2 promoter, the protein expression level for CDH production is far behind that achievable by native CBH II. Different from the synergism observed with lactose as the carbon source, the most significant synergism between CDH and cellulase components was observed in FPA catalysis rather than -glucosidase.

Mentioned biomedical entities:

Genes: Ace2, Actin, BGL1, CEL1B, CEL3A, CEL3B, Cellulase, EG II, FPA, MA, XYR1, bgl1, cbh1, cbh2, cdh, cel1a, cellulase

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Amicon Ultra-15, EC, EC 3.2.1.4, GH3, Phanerochaete chrysosporium, Phanerochaete chrysosporium), Pichia, Pichia clones, Pichia pastoris GS115, Plasmid, RUT-C3, Tm-BglA, Trichoderma transformants, cell, cells, pastoris GS115, pastoris KM71, wild-type

Organisms: bovine serum albumin

Tissues: Amicon Ultra-15, Pre-grown mycelia

Pathways: N/A

Query Result 16/25

Characterization and Strain Improvement of a Hypercellulytic Variant, Trichoderma reesei SN1, by Genetic Engineering for Optimized Cellulase Production in Biomass Conversion Improvement

Qian, Yuanchao; Zhong, Lixia; Hou, Yunhua; Qu, Yinbo; Zhong, Yaohua

Frontiers in Microbiology

Year: 2016 PMID: 27621727 doi: 10.3389/fmicb.2016.01349 Citations: 11 Score: 9.87

Keywords hits: xyr1: , ace3: , cbh1: 7, cellulase: 54, reesei: 61

Abstract:

The filamentous fungus Trichoderma reesei is a widely used strain for cellulolytic enzyme production. A hypercellulolytic T. reesei variant SN1 was identified in this study and found to be different from the well-known cellulase producers QM9414 and RUT-C3. The cellulose-degrading enzymes of T. reesei SN1 show higher endoglucanase (EG) activity but lower -glucosidase (BGL) activity than those of the others. A uracil auxotroph strain, SP4, was constructed by pyr4 deletion in SN1 to improve transformation efficiency. The BGL1-encoding gene bgl1 under the control of a modified cbh1 promoter was overexpressed in SP4. A transformant, SPB2, with four additional copies of bgl1 exhibited a 17.1-fold increase in BGL activity and a 3.% increase in filter paper activity. Saccharification of corncob residues with crude enzyme showed that the glucose yield of SPB2 is 65.% higher than that of SP4. These results reveal the feasibility of strain improvement through the development of an efficient genetic transformation platform to construct a balanced cellulase system for biomass conversion.

Summary:

As shown in. reesei SN1 produced a total cellulase activity comparable to that of QM9414 but exhibited a notably higher EG activity than QM9414 and RUT-C3. Comparison of CBH and BGL activities showed that SN1 also produced a CBH activity similar to that of QM9414 but had a much reduced BGL activity. Detection of -glucosidase activity on the CMC-esculin plate in T. reesei SN1, QM9414 and RUT-C3 at 3C for 24 h. The FPA and EG activities of the T. reesei SN1, QM9414 and RUT-C3, were measured after being induced in cellulase fermentation medium for 5 days. In accordance with this result, the BGL activities of SPB1 and SPB2 measured with p NPG as a substrate were 4.7 and 17.1 times higher than that of SP4, respectively. The ratio of BGL activity to FPA in SPB2 reached 2.8, which was much higher than that in the parental strain.

Mentioned biomedical entities:

Genes: BGL1, CBH, Cellulase, EG, FPA, SP4, SPB2, Y2, bgl1, cbh1, cellulase, cre1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, Culture ConditionsTrichoderma reesei QM9414, EG, Four-Copy cbh1, Penicillium, RUT-C3, TA, b Strains, cell, cells, cellular, colonies, lignin, mycelium, seed cultures, spore, transformant SPB1

Organisms: E. coli hygromycin B phosphotransferase, Escherichia coli, pyr4 locus

Tissues: pulp

Pathways: N/A

Query Result 17/25

Enhancing cellulase and hemicellulase production by genetic modification of the carbon catabolite repressor gene, creA, in Acremonium cellulolyticus

Fujii, Tatsuya; Inoue, Hiroyuki; Ishikawa, Kazuhiko

AMB Express

Year: 2013 PMID: 24360128 doi: 10.1186/2191-0855-3-73 Citations: 14 Score: 9.57

Keywords hits: xyr1: 5, ace3: , cbh1: , cellulase: 71, reesei: 2

Abstract:

Acremonium cellulolyticus is one of several fungi that offer promise as an alternative to Trichoderma reesei for use in industrial cellulase production. However, the mechanism of cellulase production has not been studied at the molecular level because adequate genetic engineering tools for use in A. cellulolyticus are lacking. In the present study, we developed a gene disruption method for A. cellulolyticus, which needs a longer homologous region length. We cloned a putative A. cellulolyticus creA gene that is highly similar to the creA genes derived from other filamentous fungi, and isolated a creA disruptant strain by using the disruption method. Growth of the creA disruptant on agar plates was slower than that of the control strain. In the wild-type strain, the CreA protein was localized in the nucleus, suggesting that the cloned gene encodes the CreA transcription factor. Cellulase and xylanase production by the creA disruptant were higher than that of the control strain at the enzyme and transcription levels. Furthermore, the creA disruptant produced cellulase and xylanase in the presence of glucose. These data suggest both that the CreA protein functions as a catabolite repressor protein, and that disruption of creA is effective for enhancing enzyme production by A. cellulolyticus.

Summary:

Because A. cellulolyticus is an industrially important fungal species, how cellulase and hemicellulase gene expression is regulated in this organism is crucial for the development of more efficient enzyme production methods. In the present study, we cloned a putative cre A gene from A. cellulolyticus that is highly similar to the cre A genes of other filamentous fungi and then isolated a recombinant strain in which the cre A gene was disrupted. Strains were cultured in medium containing 5 g/L of cellulose, or 1 g/L of cellulose and 4 g/L of glucose. Gene expression was much higher in YDCre than in YPyr F, which is consistent with the result showing that cellulase and xylanase enzymatic activity are higher in YDCre than in YPyr F. When the strains were cultured in medium containing cellulose and glucose, no expression was detected in YPyr F for any of the genes analyzed. Enzyme production in a T. reesei, cre A-knockout strain was shown to be higher than in the parental strain under certain conditions, which is consistent with our results described above. In other filamentous fungi, Cre A protein regulates gene expression by binding to the promoter region.

Mentioned biomedical entities:

Genes: ACE1, ACE2, Ace1, Ace2, AraR, AreA, BglR, Cellulase, CreA, CreA protein, XYR1, XlnR, Xyr1, cel5A, cel7B, cellulase, creA, gpdA, pyrF

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: A. cellulolyticus, A. cellulolyticus Y-94, Cellular, MM, Penicillium, Penicillium echinulatum, Trichoderma koningii, YDCre cells, cells, cellulose, cellulose-

Organisms: A. cellulolyticus, A. niger, YPyrF creA

Tissues: Fiber Sales

Pathways: N/A

Query Result 18/25

Effect of highly branched hyphal morphology on the enhanced production of cellulase in Trichoderma reesei DES-15

He, Ronglin; Li, Chen; Ma, Lijuan; Zhang, Dongyuan; Chen, Shulin

3 Biotech
Year: 2016 PMID: 28330286 doi: 10.1007/s13205-016-0516-5 Citations: 13 Score: 9.28
Keywords hits: xyr1: 11, ace3: , cbh1: 5, cellulase: 42, reesei: 9

Abstract:
The morphology of Trichoderma reesei is a vitally important factor for cellulase productivity. This study investigated the effect of hyphal morphology on cellulase production in the hyper-cellulolytic mutant, T. reesei DES-15. With a distinct morphology, T. reesei DES-15 was obtained through Diethyl sulfite (DES) mutagenesis. The hyphal morphology of DES-15 batch-cultured in a 5-L fermentor was significantly shorter and more branched than the parental strain RUT C3. The cellulase production of DES-15 during batch fermentation was 66% greater than that of RUT C3 when cultured the same conditions. DES-15 secreted nearly 5% more protein than RUT C3. The gene expression level of a set of genes (cla4, spa2, ras2, ras1, rhoA, cdc42, and racA) known to be involved in hyphae growth and hyphal branching was measured by quantitative real-time PCR. The transcriptional analysis of these genes demonstrated that a decrease in gene expressions might contribute to the increased hyphal branching seen in DES-15. These results indicated that the highly branching hyphae in DES-15 resulted in increased cellulase production, suggesting that DES-15 may be a good candidate for use in the large-scale production of cellulase.

Summary:
Numerous studies have demonstrated that changing important parameters during the fermentation process, such as the initial spore concentration, medium composition, p H value, temperature, and agitation, can control fungal morphology (Gibbs et al. 2 Papagianni 24 Ferreira et al. 29 Peciulyte et al. Nevertheless, obtaining a strain with optimal morphology in submerged cultures is an important factor for the production of cellulase by T. reesei. In this study, we utilized a hyper-cellulolytic mutant with a unique morphology, where the hyphae appeared to be shorter and more branched during fermentation. The amount of reducing sugars produced was determined following a 6 min culture, where 5 L of an enzyme solution is mixed with 1 m L of citrate buffer in the presence of 5 mg of Whatman No. 1), the cellulase activity for the parent strain RUT C3 was determined to be 7.11 FPU/m L, while the cellulase activity of one hyper-producing cellulase mutant, DES-15, was determined to be 11.86 FPU/m L, a 66 % increase in the overall cellulase activity. As this is an initial screening, data are representative for a single liquid fermentation only, b: bSubmerged cultivation in a fermentor is mostly used in the large-scale production of cellulase, as many parameters are easier to control in fermentors compared to flasks during large-scale production.

Mentioned biomedical entities:

Genes: Cdc42, Cla4, FPA, IPP, RUT, RacA, Ras1, Ras2, RhoA, Spa2, XYR1, bgl1, cbh1, cdc42, cellulase, gpd1, racA, rhoA, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Chang, Erlenmeyer flasks, Trichoderma, b, bBoth, cell, clones, colonies, conidia, cultures, hypha, spore

Organisms: C. Li, human

Tissues: lateral

Pathways: N/A

Query Result 19/25

Constitutive overexpression of cellobiohydrolase 2 in Trichoderma reesei reveals its ability to initiate cellulose degradation

Wang, Yubo; Ren, Meibin; Wang, Yifan; Wang, Lu; Liu, Hong; Shi, Mei; Zhong, Yaohua

Engineering Microbiology

Year: 2023 PMID: 39628517 doi: 10.1016/j.engmic.2022.100059 Citations: 2 Score: 8.91

Keywords hits: xyr1: 1, ace3: , cbh1: 21, cellulase: 44, reesei: 49

Abstract:

Overexpression of cbh2 with different constitutive promoters in Trichoderma reesei.Cellobiohydrolase 2 promotes the early-stage expression of major cellulase genes.Cellobiohydrolase 2 contributes to the initiation of cellulose degradation.

Summary:

Thus, two constitutive promoters, Pgpd1 and Prp S3, were used to drive the overexpression of CBH2 in this study to further examine the effect of CBH2 on cellulase induction and cellulose degradation. First, the cbh2 expression cassettes, which contain the promoter gpd1 or rp S3, the cbh2 gene, and the trp C terminator, were constructed using Double-joint PCR. The overexpression strains with Pgpd1 as a promoter: Qgc25, Qgc218, and Qgc221. The overexpression strains with Prp S3 as a promoter: Qrc234, Qrc232, Qrc24, and the parental strain T. reesei QM9414.2 The high level of transcription of cbh2 under glucose conditions T. The results indicated that the transcriptional levels of cbh2 in Qgc225 and Qrc24 were 3687-fold and 1868-fold, respectively, compared with that of the parental strain QM9414 (Fig. Overexpression of CBH2 triggered the induction of the major cellulase genes to be significantly elevated, which is probably attributable to the CBH2s ability to begin the initial attack on cellulose to manufacture the cellulase inducers 33,52,53 and causes the cellulose surface to be heavily destroyed during the early stage of cellulose degradation. Altogether, the results of this study will contribute to a better understanding of CBH2 function in the process of cellulosic material degradation by T. reesei, which has essential practical implications for the optimization of cellulase cocktails to further improve the saccharification efficiency.

Mentioned biomedical entities:

Genes: AOX1, CBH, CBH1, CBH2, CRE1, Cellulase, FPA, GAP, GPD, RPS3, actin, cbh1, cbh2, cellulase, eg1, gpd1, rpS3

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: 5-mL Erlenmeyer flasks containing 1-mL MM , Bio-rad DC, C, EC, EC 3.2.1.4, MML, Pichia pastoris, Qrc24, cbh2-probe-R, cell, colonies, conidia, corn plasm, microcrystalline cellulose.21T. reesei QM9414gpd1, pastoris, seed medium, solid

Organisms: N/A

Tissues: smooth

Pathways: N/A

Query Result 20/25

Factors regulating cellulolytic gene expression in filamentous fungi: an overview

Mattam, Anu Jose; Chaudhari, Yogesh Babasaheb; Velankar, Harshad Ravindra

Microbial Cell Factories

Year: 2022 PMID: 35317826 doi: 10.1186/s12934-022-01764-x Citations: 0 Score: 8.91

Keywords hits: xyr1: 68, ace3: 24, cbh1: 17, cellulase: 2, reesei: 15

Abstract:

The growing demand for biofuels such as bioethanol has led to the need for identifying alternative feedstock instead of conventional substrates like molasses, etc. Lignocellulosic biomass is a relatively inexpensive feedstock that is available in abundance, however, its conversion to bioethanol involves a multistep process with different unit operations such as size reduction, pretreatment, saccharification, fermentation, distillation, etc. The saccharification or enzymatic hydrolysis of cellulose to glucose involves a complex family of enzymes called cellulases that are usually fungal in origin. Cellulose hydrolysis requires the synergistic action of several classes of enzymes, and achieving the optimum secretion of these simultaneously remains a challenge. The expression of fungal cellulases is controlled by an intricate network of transcription factors and sugar transporters. Several genetic engineering efforts have been undertaken to modulate the expression of cellulolytic genes, as well as their regulators. This review, therefore, focuses on the molecular mechanism of action of these transcription factors and their effect on the expression of cellulases and hemicellulases.

Summary:

In another study, the overexpression of the Xln R gene with an A871 V point mutation and Clr B, along with the deletion of cre A resulted in a strain with 8.9- and 51.5-fold higher production of cellulases and xylanases, respectively 57. In fact, the overexpression of Ace4 led to a 22% increase in the expression of the major cellulase genes, and this effect was mediated in an Ace3-dependent manner 62. The deletion of Ace3 resulted in negligible expression of cbh1, cbh2, egl1, bgl1, and xyn3 in the recombinant strain as compared to the parental strain 31. It has been observed that Xpp1 regulates the transcription of hemicellulase genes only in the later stages of growth and no significant difference in xylanase activities was observed in an xpp1-disrupted strain or its parent strain, in the initial 72 h, reconfirming its role as a repressor of secondary metabolism 112. Sxl RThe Sxl R transcription factor was identified due to its role in inhibiting the expression of the major xylanases such as xyn1, xyn2, xyn5, etc. As expected, the overexpression of the Ctf1 gene under the constitutive pdc1 promoter led to significant repression of cellulase genes 146. In another study, Rce2, a protein recently identified by a pull-down and mass spectrometry analysis was found to have similar binding sites in the promoter regions of target genes as Ace3, and its overexpression led to reduced cellulase expression i. e. Rce2 acted antagonistic to Ace3 in T. reesei with regards to cellulase induction and led to repressed cellulase and hemicellulase expression 147. The calcineurin-responsive zinc finger transcription factor 1 or Crz1 was identified in T. reesei by gene disruption 148. Despite significant progress made in studies related to the production of cellulases, the influence of sugar transporters on the degradation of cellulose or lignocellulosic biomass has not been investigated in detail. The deletion of the MFS protein Stp1 that transports cellobiose in T. reesei, resulted in the repression of both cellulase and hemicellulase genes, in presence of Avicel however, no effect on cellulase expression was observed when the strain was grown in cellobiose.

Mentioned biomedical entities:

Genes: Ace1, Ace2, Ace3, AceI, Are1, AreA, Azf1, B, BglR, CYC8, Cdt1, CdtC, Clr1, Clr2, Cre1, CreA, CreB, CreC, Crt1, Crz1, Ctf1, GH, GHs, Gal11, HapB, LaeA, MFS, MFS protein, Pac1, PacC, PalA, Stp1, TUP1, TacA, Vel2, Vel3, VelB, Vib1, XlnR, Xyr1, ace2, amdS, bgl1, bgl2, cbh1, cbh2, cellulase, cre, cre1, eg1, pac1, pal, pdc, tubB, xlnR, xyn1, xyn2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: EC, F, M2 transformant, Penicillium, QM6a, Trichoderma, Trichoderma reesei, cell, cellulases, cerevisiae, colonies, wheat, wheat bran

Organisms: A. niger CreA, A. niger XlnR, S. cerevisiae, S. cerevisiae NDT8

Tissues: N/A

Pathways: N/A

Query Result 21/25

The effects of disruption of phosphoglucose isomerase gene on carbon utilisation and cellulase production in Trichoderma reesei Rut-C3

Limn, M Carmen; Pakula, Tiina; Saloheimo, Markku; Penttil, Merja

Microbial Cell Factories

Year: 2011 PMID: 21609467 doi: 10.1186/1475-2859-10-40 Citations: 6 Score: 8.78

Keywords hits: xyr1: 18, ace3: , cbh1: 1, cellulase: 76, reesei: 41

Abstract:
Cellulase and hemicellulase genes in the fungus Trichoderma reesei are repressed by glucose and induced by lactose. Regulation of the cellulase genes is mediated by the repressor CRE1 and the activator XYR1. T. reesei strain Rut-C3 is a hypercellulolytic mutant, obtained from the natural strain QM6a, that has a truncated version of the catabolite repressor gene, cre1. It has been previously shown that bacterial mutants lacking phosphoglucose isomerase (PGI) produce more nucleotide precursors and amino acids. PGI catalyzes the second step of glycolysis, the formation of fructose-6-P from glucose-6-P.

Summary:
Cellulase production in the mutants started slightly earlier in MM with fructose and glucose than in MM with only glucose but maximal production was similar in both media. Cellulase activity of cre1+pgi1 mutants was measured both in MM with glucose and fructose and in MM with lactose and fructose, after 7 days of culture. In conclusion, deletion of gene pgi1, both in a cre1-1 and a cre1+ background, caused an increase in cellulase activity in medium with glucose and a drop of the activity in medium with lactose. Table 1 Cellulase activity under repressing and inducing conditions Strains Glu and Fru Lac and Fru QM-6a .15.21 .43 Rut C-3.61 .55.95 .68 P41.44 .63.48 .66 CP26.29 .171.22 .41 CP28.33 .172.66 .26 Activity was measured from samples taken after 7 days of incubation in MM with glucose and fructose and MM with lactose and fructose. A striking phenotype of the pgi1 mutants was the pellet-like growth that was observed in MM with glucose and in MM with glucose and fructose. As observed under a microscope, the mycelial morphology of pgi1 mutants in media with glucose is reminiscent of the morphology of mutants affected in the production of cell wall. In addition, the lower production of extracellular proteins in the pgi1 mutants in MM with lactose and fructose could also be explained by the slow hydrolysis of lactose, due to a low -galactosidase activity. Sugar utilisation Glucose consumption rate of the pgi1 strains was reduced, in comparison with the parental strain Rut-C3, when grown in MM with glucose.

Mentioned biomedical entities:

Genes: ACE1, ACE2, B, CP26, CP28, CRE1, Cellulase, GDH2, GPDH, MP, NADPH, P4, P58, PGI, PGI1, TP, XYL1, XYR1, cbh1, cellulase, cre1, xyl1, xyn1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, Cell extracts, E. coli , F and 7, F) Conidia, F) MM, F, G, Figure 8A).In cultures, Fusarium oxysporum , Hypocrea jecorinaFEBS, Kluyveromyces lactis rag2, L6, MM, Monosporic cultures, QM6a, Rut-C3, Rut-C3 mycelia, RutC-3, Saccaromyces, Saccharomyces, Saccharomyces cerevisiae strainsFEMS, Saccharomyces cerevisiaeFEMS Microbiol Lett19931632332371.1111/j.1574, TOP1 E. coli, Trichoderma, Trichoderma harzianum transformants, Trichoderma strains, basal cells, cell, cell extracts, cell extractsMycelial samples, cell wall, cells, cellular, cerevisiae, clone, clones, cosmid clone, culture filtrates, cultures, hygromycin, left, liquid, organelles, plant cell wall, plasmid, solid

Organisms: E. coli, Escherichia coli, H and, H and 7, RNA samples, S. cerevisiae, roots

Tissues: vascular wilt, pgi1 colonies, pgi1 mutant

Pathways: N/A

Query Result 22/25

Improved Production of Majority Cellulases in Trichoderma reesei by Integration of cbh1 Gene From Chaetomium thermophilum

Jiang, Xianzhang; Du, Jiawen; He, Ruonan; Zhang, Zhengying; Qi, Feng; Huang, Jianzhong; Qin, Lina

Frontiers in Microbiology

Year: 2020 PMID: 32765463 doi: 10.3389/fmicb.2020.01633 Citations: 0 Score: 8.54

Keywords hits: xyr1: 1, ace3: 6, cbh1: 29, cellulase: 7, reesei: 117

Abstract:
Lignocellulose is an abundant waste resource and has been considered as a promising material for production of biofuels or other valuable bio-products. Currently, one of the major bottlenecks in the economic utilization of lignocellulosic materials is the cost-efficiency of converting lignocellulose into soluble sugars for fermentation. One way to address this problem is to seek superior lignocellulose degradation enzymes or further improve current production yields of lignocellulases. In the present study, the lignocellulose degradation capacity of a thermophilic fungus Chaetomium thermophilum was firstly evaluated and compared to that of the biotechnological workhorse Trichoderma reesei. The data demonstrated that compared to T. reesei, C. thermophilum displayed substantially higher cellulose-utilizing efficiency with relatively lower production of cellulases, indicating that better cellulases might exist in C. thermophilum. Comparison of the protein secretome between C. thermophilum and T. reesei showed that the secreted protein categories were quite different in these two species. In addition, to prove that cellulases in C. thermophilum had better enzymatic properties, the major cellulase cellobiohydrolase I (CBH1) from C. thermophilum and T. reesei were firstly characterized, respectively. The data showed that the specific activity of C. thermophilum CBH1 was about 4.5-fold higher than T. reesei CBH1 in a wide range of temperatures and pH. To explore whether increasing CBH1 activity in T. reesei could contribute to improving the overall cellulose-utilizing efficiency of T. reesei, T. reesei cbh1 gene was replaced with C. thermophilum cbh1 gene by integration of C. thermophilum cbh1 gene into T. reesei cbh1 gene locus. The data surprisingly showed that this gene replacement not only increased the cellobiohydrolase activities by around 4.1-fold, but also resulted in stronger induction of other cellulases genes, which caused the filter paper activities, Azo-CMC activities and -glucosidase activities increased by about 2.2, 1.9, and 2.3-fold, respectively. The study here not only provided new resources of superior cellulases genes and new strategy to improve the cellulase production in T. reesei, but also contribute to opening the path for fundamental research on C. thermophilum.

Summary:
One way to overcome these drawbacks is the introduction of lignocellulases encoding genes from thermophilic fungi into mesophilic species, in which a variety of genetic tools have been developed, to combine the advantages of mesophilic and thermophilic properties. The soft-rot fungus Trichoderma reesei is a model organism for plant biomass degradation and widely used in industry for the production of cellulases and xylanases due to the large capacity of hydrolytic enzymes secretion. Further analysis of the extracellular protein levels in the culture supernatants of C. thermophilum and T. reesei showed that under Avicel, the levels of secreted proteins in C. thermophilum were slightly higher than that in T. reesei in the first 3 days. Shown in is one of the three represents. Chaetomium thermophilum Contained More Extracellular Protein Categories Compared to T. reesei To further investigate the mechanism behind the higher cellulose degradation efficiency in C. thermophilum, the supernatants of C. thermophilum and T. reesei cultures under Avicel were analyzed by SDS-PAGE. Considering that the certain oxidants could make lignocellulose more susceptible for enzymatic degradation, the existence of such amounts of oxidoreductases in the supernatants of C. thermophilum culture implied that in addition to contain higher efficient cellulases, C. thermophilum might display a different mode of action to degrade lignocelluloses. In T. reesei, it has been generally believed that -glucosidase is a barrier for its cellulose degradation capacity because the extracellular -glucosidase comprises only about 1% of the total T. reesei cellulases complex, based on which, many efforts have been conducted to increase enzyme efficiency of hydrolyzing cellulosic substrates by increasing -glucosidase amount in the cellulases complex from T. reesei. The results that the FPA activities, the Azo-CMC activities and -glucosidase activities in Tr-Ctcbh1 strains were significantly higher than those in Cpyr4 strains, and besides, the transcriptional levels of cbh2, eg1, eg2, and bgl1 genes were also higher in Tr-Ctcbh1 strains, which further proved that the induction and expression of most cellulases genes in T. reesei could be improved by only enhancing CBH1 activities. The increased expression levels of xyr1, cre1, and ace3 in Tr-Ctcbh1 strains suggested that an extra inducer might be generated in T. reesei when response to cellulose with more cellobiohydrolases in its cellulases system.

Mentioned biomedical entities:

Genes: CBH, CBH1, FPA, MUC, ace1, ace2, actin, bgl1, cbh1, cbh2, cellulase, cre1, eg1, pyr4, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Avicel cultures, Azo-CMC, B, C, Chaetomium, ConditionsEscherichia coli Trans1-T1, Culture, FgDNA-P, Liquid Chromatography-Mass

SpectrometryThe, Penicillium species, TFs, Tr-Ctcbh1, Tr-Ctcbh1 strains, Tr-Ctcbh1 transformants, Tr-cCcbh1, Tr-cCcbh1 cultures, Tr-cCcbh1 transformants, Tr-cTcbh1, Tr-cTcbh1 strains, Tr-cTcbh1 transformants, Tr-cbh1, YPD, b SDS-PAGE, cellulases, cellulose cultures, column, plasmid, spore

Organisms: C.

Tissues: Tr-cCcbh1 strains, Tu6ku7 strains, wet mycelia

Pathways: N/A

Query Result 23/25

Genus-wide analysis of Trichoderma antagonism toward Pythium and Globisporangium plant pathogens and the contribution of cellulases to the antagonism

Chen, Siqiao; Daly, Paul; Anjago, Wilfred Mabeche; Wang, Rong; Zhao, Yishen; Wen, Xian; Zhou, Dongmei; Deng, Sheng; Lin, Xisha; Voglmeir, Josef; Cai, Feng; Shen, Qirong; Druzhinina, Irina S.; Wei, Lihui; Johnson, Karyn N.

Applied and Environmental Microbiology

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Keywords hits: xyr1: 41, ace3: , cbh1: , cellulase: 94, reesei: 6

Abstract:
Parasitism is an important lifestyle in the Trichoderma genus but has not been studied in a genus-wide way toward Pythium and Globisporangium hosts. Our approach screened a genus-wide set of 3 Trichoderma species in dual culture assays with two soil-borne Pythium and three Globisporangium plant-parasitic species and used exo-proteomic analyses, with the aim to correlate Trichoderma antagonism with potential strategies for attacking Pythium and Globisporangium. The Trichoderma spp. showed a wide range of antagonism from strong to weak, but the same Trichoderma strain showed similar levels toward all the Pythium and Globisporangium species. The Trichoderma enzymes from strong (Trichoderma asperellum, Trichoderma atroviride, and Trichoderma virens), moderate (Trichoderma cf. guizhouense and Trichoderma reesei), and weak (Trichoderma parepimyces) antagonists were induced by the autoclaved mycelia of one of the screened Pythium species, Pythium myriotylum. The variable proportions of putative cellulases, proteases, and redox enzymes suggested diverse as well as shared strategies amongst the antagonists. There was a partial positive correlation between antagonism from microscopy and the cellulase activity induced by autoclaved P. myriotylum mycelia in different Trichoderma species. The deletion of the cellulase transcriptional activator XYR1 in T. reesei led to lower antagonism toward Pythium and Globisporangium. The antagonism of Pythium and Globisporangium appears to be a generic property of Trichoderma as most of the Trichoderma species were at least moderately antagonistic. While a role for cellulases in the antagonism was uncovered, cellulases did not appear to make a major contribution to T. reesei antagonism, and other factors are also likely contributing.

Summary:
Secreted Trichoderma proteins, such as cell wall-degrading enzymes, can be an important factor in antagonism, and the autoclaved P. myriotylum mycelia were used to investigate which secreted proteins were produced by these Trichoderma species used for the microscopic analysis that had different levels of antagonism toward the Pythium and Globisporangium hosts. To overcome these, the cellulase activity in plate-based cultures was measured using single cultures, and a Trichoderma mutant with reduced cellulase activity was used in confrontation assays with Pythium and Globisporangium hosts. Partial correlation of cellulase activity in plate cultures with the level of antagonism AZCL-cellulose is a blue-dyed chromogenic cellulose polymer, whereby the hydrolysis of the cellulose is detected by the intensity and spread of the blue dye in agar plates. P. myriotylum was re-isolated from the dead seedlings infected by P. myriotylum, and T. asperellum was re-isolated from the disease-protected seedlings that were inoculated with both P. myriotylum and T. asperellum, and the identification was confirmed by sequencing of the ITS region. DISCUSSIONHere, we showed that antagonism toward Pythium and Globisporangium appears to be a generic property of Trichoderma, various antagonism-related proteins are induced by P. myriotylum mycelium, and cellulases are a contributing factor to the antagonism of Pythium and Globisporangium species. The ancestral lifestyle of Trichoderma is considered to be mycoparasitism, and later, Trichoderma evolved to degrade plant cell walls via horizontal gene transfer of plant cell wall-degrading enzymes including cellulases. Since the tested xyr1 mutant in our study was in a moderate antagonist, for further investigating the contribution of cellulases to antagonism, it would be of interest to delete xyr1 in strong Trichoderma antagonists. As the exo-proteomic analysis and orthogroup analysis stated in the result section, there was a positive correlation to the weakest antagonism of T. parepimyces in plate confrontation and the absence of GH7 putative cellobiohydrolases which appeared to be one of the main cellulases present that could be involved in degrading the cellulose polymer found in the Pythium and Globisporangium cell wall. To prepare the inoculum used on the AZCL-cellulose plates, the freshly harvested mature spores from the selected Trichoderma strains were inoculated into a minimal medium containing 1% D-fructose and .1% peptone and incubated at 2 RPM and 28C in the dark for up to 2 h. Then 1 13 pre-germinated spores were inoculated onto a corner of each AZCL-cellulose plate and were incubated at 28C in the dark for up to 5 d. The images of the Trichoderma colonies with the dye-releasing zone were taken by a Canon EOS R6 camera with back illumination from a lightbox. Calcofluor white staining and microscopy of confrontations in situ For better hyphal observations in situ of dual culture in confrontation assay, the 35 mm microscopic dishes were used in this study for microscopic investigation.

Mentioned biomedical entities:

Genes: Cellulase, XYR1, cellulase, cre1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, C18 HyperClone 5, CBS, Crop, G., Globisporangium cell wall, MR2, Pythium, Pythium aphanidermatum cells, Pythium cell, RUT-C3, SEs, TU, TU-6, Trichoderma, Trichoderma cell wall-degrading, Trichoderma colonies, Trichoderma cultures, Trichoderma enzymesFreshly, Trichoderma longibrachiatum, Trichoderma strains, Trichoderma-Pythium, black lines, cell, cell wall, cell wall-degrading enzymes, cell walls, cells, columns, cultures, endo-cellulase, endophyte, freeze-dried, hyphal cell wall, layer, line, mycelium, oomycete cell walls, oospore, plant cell wall-degrading enzymes, shake-flask cultures, soil-borne, soil-borne Pythium, solid cultures, spore, stem, virens, zone

Organisms: G. ultimum cell wall, Koch, bovine serum albumin, roots, tobacco

Tissues: Trichoderma hyphae, leaf sheath collar, tissue

Pathways: N/A

Query Result 24/25

From induction to secretion: a complicated route for cellulase production in Trichoderma reesei

Yan, Su; Xu, Yan; Yu, Xiao-Wei

Bioresources and Bioprocessing

Year: 2021 PMID: 38650205 doi: 10.1186/s40643-021-00461-8 Citations: 0 Score: 8.32

Keywords hits: xyr1: 26, ace3: 1, cbh1: , cellulase: 21, reesei: 77

Abstract:
The filamentous fungus Trichoderma reesei has been widely used for cellulase production that has extensive applications in green and sustainable development. Increasing costs and depletion of fossil fuels provoke the demand for hyper-cellulase production in this cellulolytic fungus. To better manipulate T. reesei for enhanced cellulase production and to lower the cost for large-scale fermentation, it is wise to have a comprehensive understanding of the crucial factors and complicated biological network of cellulase production that could provide new perspectives for further exploration and modification. In this review, we summarize recent progress and give an overview of the cellular process of cellulase production in T. reesei, including the carbon source-dependent cellulase induction, complicated transcriptional regulation network, and efficient protein assembly and trafficking. Among that, the key factors involved in cellulase production were emphasized, shedding light on potential perspectives for further engineering.

Summary:
1 Overview of cellulase induction in T. reesei. However, the critical factors are gradually investigated, which gives a clue for further research, and it values a lot to engineer cellulase production on lactose which would significantly lower the cost for cellulase production. Sugar transporters are engaged in cellulase induction In cellulolytic fungus T. reesei, sugar transporters are essential for the perception of cellulose and the uptake of soluble disaccharides, which is critical for efficient cellulase induction. A binding competition between RCE2 and ACE3 was also observed, indicating a synergetic regulation among RCE2 and other transcription factors. Other factors involved in transcriptional regulation Aside from these specific transcription factors, other factors involved in secondary metabolism, signal transduction and chromatin status regulation also participate in the transcriptional modification of cellulase production in different aspects (Fig. The mitogen-activated protein kinase pathway, which is ubiquitous in eukaryotes and is involved in biological regulation, could also participate in the regulation of cellulase expression through the phosphorylation of transcription factors by TMK3 in T. reesei. Although only a few research have been conducted in T. reesei, the characterized mechanism in other fungi could also give a clue for further engineering in T. reesei.

Mentioned biomedical entities:
Genes: ACE2, ACE3, ALP, ARE1, BiP1, CCR, CEL1B, CEL3A, CEL7A, CLP1, CLR1, CRE1, CRT1, CRZ1, CTF1, Cellulase, CrzA, ER, GNA-3, HAC1, IRE1, Kar2, LAE1, LaeA, MAPK, PAC1, PDI1, PKA, PacC, RCE1, Rho, SAR1, SEC16, SLY1, SNC2, SSO1, STP1, STR1, STR3, VEL2, VIB1, VelB, VelC, XYR1, YPR1, bgl1, cAMP, cellulase, cre1, p24, pdi1, rac1, sar1, swo1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: AP-3, GH1, GH3, Golgi, Pichia pastoris, TFs, b, cell, cell plasma membrane, cellular, cerevisiae, histones, lines, photoreceptors

Organisms: A. niger, A. oryzae

Tissues: hypha

Pathways: N/A

Query Result 25/25

The protein methyltransferase TrSAM inhibits cellulase gene expression by interacting with the negative regulator ACE1 in Trichoderma reesei

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Communications Biology

Year: 2024 PMID: 38548869 doi: 10.1038/s42003-024-06072-1 Citations: 0 Score: 8.18

Keywords hits: xyr1: 34, ace3: 2, cbh1: 34, cellulase: 58, reesei: 48

Abstract:

Protein methylation is a commonly posttranslational modification of transcriptional regulators to fine-tune protein function, however, whether this regulation strategy participates in the regulation of lignocellulase synthesis and secretion in Trichoderma reesei remains unexplored. Here, a putative protein methyltransferase (TrSAM) is screened from a T. reesei mutant with the ability to express heterologous -glucosidase efficiently even under glucose repression. The deletion of its encoding gene trsam causes a significant increase of cellulase activities in all tested T. reesei strains, including transformants of expressing heterologous genes using cbh1 promotor. Further investigation confirms that TrSAM interacts with the cellulase negative regulator ACE1 via its amino acid residue Arg383, which causes a decrease in the ACE1-DNA binding affinity. The enzyme activity of a T. reesei strain harboring ACE1R383Q increases by 85.8%, whereas that of the strains with trsam or ace1 deletion increases by more than 1%. By contrast, the strain with ACE1R383K shows no difference to the parent strain. Taken together, our results demonstrate that TrSAM plays an important role in regulating the expression of cellulase and heterologous proteins initiated by cbh1 promotor through interacting with ACE1R383. Elimination and mutation of TrSAM and its downstream ACE1 alleviate the carbon catabolite repression (CCR) in expressing cellulase and heterologous protein in varying degrees. This provides a new solution for the exquisite modification of T. reesei chassis.

Summary:

This confirmed that the deletion of trsam contributed to the increase of p NPGase activity in atbg-U1. When the methyltransferase-encoding gene trsam was also deleted in T. reesei QM6a and its derivatives via Agrobacterium-mediated transformation, the FPAs of the respective deletion mutants increased by 154%, 18%, and 32%, respectively. To investigate the function of Tr SAM in the regulation of cellulase production, interactions between Tr SAM and widely known cellulase-related regulators were analyzed using the yeast two-hybrid system. The interaction between Tr SAM-AD and ACE1-BD promoted yeast growth in the absence of histidine and adenine and resulted in X--gal activity, whereas the interaction between Tr SAM-AD and Lae1-BD or other fused regulators did not. After the interactional verification of Tr SAM and ACE1 or the mutated ACE1 by in vivo and in vitro assays, we concluded that the putative methyltransferase Tr SAM regulated cellulase production likely by the interaction with ACE1 at R383, especially under CCR condition. The amino acid residue R383 is crucial for ACE1 to bind the promoters of its target genes and thus results in the repression of cellulase gene expression To further explore the mechanism involved in the phenomenon that Tr SAM regulated cellulase expression, which was potentially related to ACE1, we tested the effects of different ACE1 mutations on enzyme activity. It has been testified that ACE1 and XYR1 may bind to the promoters of xyr1, cbh1 and other cellulase or hemicellulase genes competitively5. In this study, we found that the methyltransferase Tr SAM participates in the repression of cellulase synthesis by the interaction with the negative regulator ACE1 and thus strengthening its binding to the promotor regions of target genes. Although it is possible that the low expression level of ACE1 under inducing condition has weakened its interaction with Tr SAM, which further impairs its repressing effect on cellulase expression, the mutation of 383 residue arginine to glutamine or deletion of trsam has been demonstrated to improve cellulase production regardless of inducing or repressing conditions (Fig.

Mentioned biomedical entities:

Genes: ACE1, ACE2, BglS, CBH1, CCR, Cre1, FPA, GAL4, GATA4, GST, Hda1, Lae1, Mig1, MyoD, PRC2, RK, Roc1, Vel1, XYR1, ace1, bglR, cbh1, cellulase, cre1, histone, p53

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: A. tumefaciens AGL1, A. tumefaciens cells, E. Coli BL21, E. coli BL21, E.coli, N-GFP-TrSAM, Penicillium oxalicum, QM6a, Rut-C3, Saccharomyces, TF, TFs, TrSAM-AD, b, cell, conidial, muscle cells, protoplast, spore, wheat bran

Organisms: E. coli

Tissues: Fuji Film, Rut-C3 conidia

Pathways: N/A